



# Mechanism Design Review

September 30th, 2024

## Liquity V2 Mechanism Design Review

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October 2024

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# 1 Introduction

In decentralized finance (DeFi), ensuring stability and efficiency in collateralized debt position (CDP) stablecoin protocols is essential for maintaining user confidence and long-term sustainability. Liquity V2, a decentralized borrowing protocol, has gained recognition for its innovative autonomous interest rate mechanism, which allows users to freely and continuously determine their own interest rates. Rates will be dynamically adjusted in response to potential redemptions to maintain system resilience, particularly when the market perceives the stablecoin as being worth less than \$1. As the DeFi landscape evolves, the challenge of adapting these mechanisms to fluctuating market conditions while preserving decentralization and minimizing risk remains critical.

This paper presents a comprehensive design review of Liquity V2, with a focus on its interest rate and redemption mechanisms, as well as the stability pool dynamics that shape borrowing demand, interest rates, and system incentives. The primary objective is to simulate and analyze the interactions between key variables—such as trove owner behavior, interest rate adjustments, and redemption fees—to evaluate the system’s robustness. Through detailed scenario analysis and stress testing, the paper aims to provide recommendations for optimizing protocol parameters, balancing user participation incentives, and mitigating adverse outcomes during periods of market stress.

## 2 Simulation Logic

### 2.1 Background

#### Stochasticity (Ornstein-Uhlenbeck Process)

The Ornstein-Uhlenbeck (OU) process is a continuous-time stochastic process widely used to model the mean-reverting behavior of financial variables, leveraging historical data perceived in the relevant field. In our instance, it is particularly useful in modeling the expected externally-priced interest rate behavior, as well as the expected external leverage demand within the system, in an effort to apply theoretical real-world scenarios with the dynamics perceived in the Liquity system with varying parameter values. We indicate that deviations from the mean are temporary and the spread will revert to its average level over time.

### 2.2 Mathematical Background

The OU process is defined by the following stochastic differential equation (SDE):

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

Where:

- $X_t$  is the variable of interest at time  $t$
- $\theta$  is the rate of mean reversion
- $\mu$  is the long-term mean level
- $\sigma$  is the volatility parameter
- $W_t$  is a Wiener process or standard Brownian motion

### 2.3 Fitting the Parameters Using Ordinary Least Squares (OLS)

To fit the parameters of the OU process, we discretize the SDE and use ordinary least squares (OLS) regression on the discretized version, formally denoted as the Euler-Maruyama method.

### Discretized Version of the OU Process

The discretized version of the OU process over short time intervals  $\Delta t$  is given by:

$$X_{t+\Delta t} = X_t + \theta(\mu - X_t)\Delta t + \sigma\sqrt{\Delta t}\epsilon_t$$

where  $\epsilon_t \sim \mathcal{N}(0, 1)$  is a n.i.i.d. random variable.

Rearranging terms, we get:

$$X_{t+\Delta t} - X_t = \theta(\mu - X_t)\Delta t + \sigma\sqrt{\Delta t}\epsilon_t$$

This can be written as:

$$\Delta X_t = \theta(\mu - X_t)\Delta t + \sigma\sqrt{\Delta t}\epsilon_t$$

where  $\Delta X_t = X_{t+\Delta t} - X_t$ .

### OLS Regression

To fit the parameters  $\theta$  and  $\mu$  using OLS, we define:

$$\Delta X_t = \alpha - \beta X_t + \eta_t$$

Where:

- $\alpha = \theta\mu\Delta t$
- $\beta = \theta\Delta t$
- $\eta_t = \sigma\sqrt{\Delta t}W_t$

We can estimate  $\alpha$  and  $\beta$  by performing OLS regression on the observed data. The parameters of the OU process can then be recovered as:

$$\begin{aligned}\theta &= \frac{\beta}{\Delta t} \\ \mu &= \frac{\alpha}{\theta\Delta t} \\ \sigma &= \sqrt{\frac{\sum \eta_t^2}{n\Delta t}}\end{aligned}$$

Where  $n$  is the number of observations.

## 3 Interest Rate Logic

### 3.1 Base Rate Derivation

The Base Rate  $\rho_t$  can be defined as the effective market-priced interest rate outside of the Liquity system at a given time  $t$ . As a result, the user-set interest rate behavior within the Liquity V2 system is generally expected to hover around  $\rho_t$ , given the limitations otherwise employed through redemption events if the rates are deviated too much from the market price, implying arbitrage.

The Base Rate  $\rho_t$  represents the effective market-priced interest rate outside of the Liquity system at a given time  $t$ . Consequently, user-set interest rates within the Liquity V2 system are generally expected to align closely with  $\rho_t$ . Significant deviations from this market rate are mitigated by redemption events, which enforce adjustments and prevent arbitrage opportunities.

### 3.2 Determining the External Market(s)

We leverage aggregate market data from Aave v2 and v3 protocols over the past year to determine the DeFi Base Rate. This includes a 14-day smoothed weighted average rate to provide a stable continuous rate that filters out short-term noise from inflows and outflows. The data is gathered from all significant stablecoin markets on Aave, with the aggregation performed as follows:

$$\rho_t = \frac{1}{T} \sum_{k=0}^{T/\Delta t} \left( \frac{\sum_{i=1}^N W_i(t - k\Delta t) R_i(t - k\Delta t)}{\sum_{i=1}^N W_i(t - k\Delta t)} \right)$$

Where  $R_i(t)$  is the interest rate of the  $i$ th stablecoin market at time  $t$ ,  $W_i(t)$  is the relative size of market  $i$  at time  $t$ ,  $N$  is the total number of markets considered,  $\Delta t$  is the time interval for smoothing (e.g., daily), and  $T$  is the smoothing period (14 days).

While DeFi protocols such as MakerDAO provide a governance-configured deterministic “savings rate,” formally known as the Dai Savings Rate (DSR), there are multiple fundamental issues in using it to define  $\rho$ :

1. **Size:** With about \$1B locked in the DSR, it is just a fraction of the cumulative deposits on the Aave Protocol, which amount to \$3B.
2. **Market-Priced Fluctuations:** Interest rates on Aave actively represent market sentiment in real time, allowing the variance to be freely modeled and realized accordingly during large spikes or draw-downs in demand.

Therefore, we leverage the weighted aggregate stablecoin borrow APY on Aave v2 and v3 as the historical data used in Ordinary Least Squares (OLS) regression, determining the optimized parameter values for the modified Ornstein-Uhlenbeck process, known as the Cox-Ingersoll-Ross (CIR) model, to model the rate accordingly.

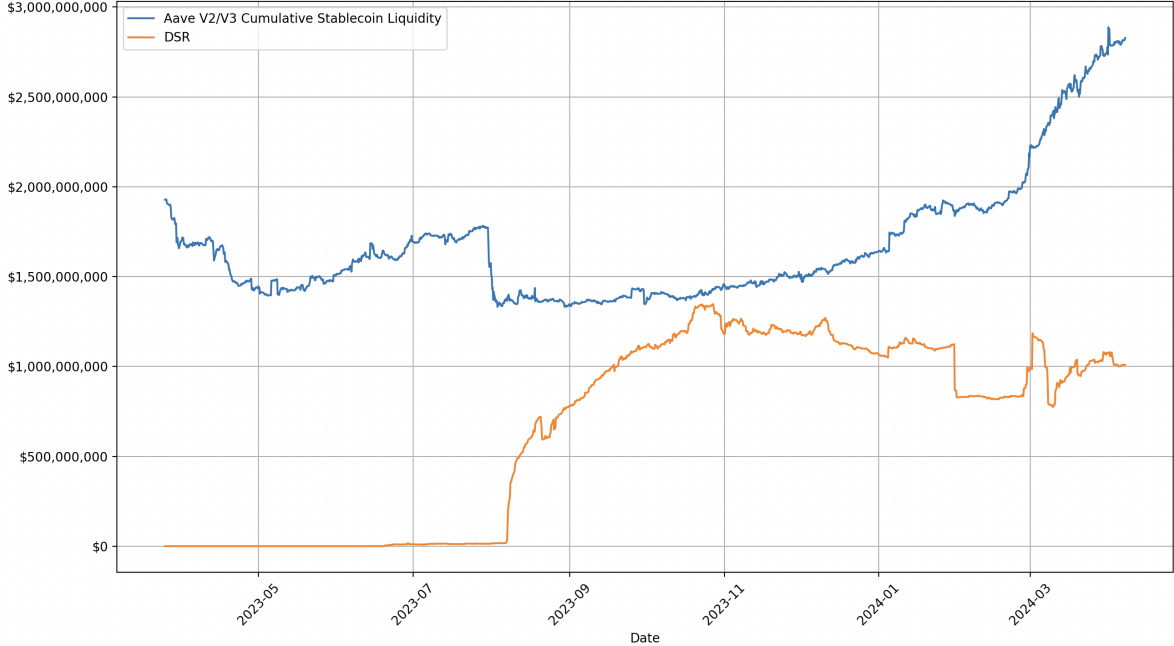


Figure 1: Aave V2/V3 Cumulative Stablecoin Liquidity and DSR over time

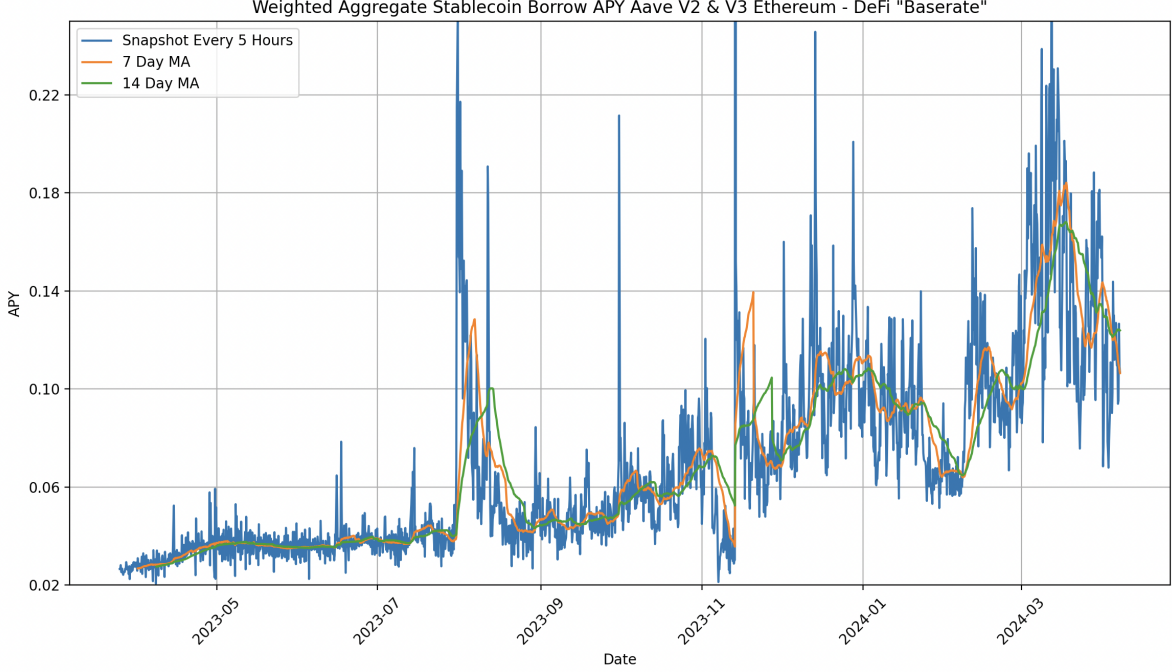


Figure 2: Weighted Aggregate Stablecoin Borrow APY Aave V2 & V3 Ethereum

### 3.3 Cox-Ingersoll-Ross Model

The Cox-Ingersoll-Ross (CIR) model is an important tool in finance for modeling interest rates. As a modified version of an Ornstein-Uhlenbeck process, the CIR model incorporates a square root term to ensure non-negativity and heteroskedasticity. The discretized version of the CIR model can be expressed using the following equation for a given time step  $\Delta\rho_t$ :

$$\rho_{t+\Delta r_t} = \rho_t + \theta(\mu - X_t)\Delta t + \sigma\sqrt{\rho_t}\sqrt{\Delta t}\epsilon_t$$

where  $\epsilon_t \sim \mathcal{N}(0, 1)$  is a standard n.i.i.d. random variable, such that:

$$\Delta\rho_t = \theta(\mu - X_t)\Delta t + \sigma\sqrt{\rho_t}\sqrt{\Delta t}\epsilon_t$$

Key characteristics of the CIR model include its ability to ensure non-negativity and heteroskedasticity in the interest rate process. The term  $\sqrt{\rho_t}$  ensures that the interest rate  $r_t$  remains non-negative, a significant advantage over models like the Vasicek model, which do not inherently prevent negative interest rates. Additionally, the presence of  $\sqrt{\rho_t}$  in the variance term means that the model accounts for changing variability in interest rates. When  $\rho_t$  is close to zero, the impact of the random shock term  $\epsilon_t$  is dampened, aligning with real-world observations where very low interest rates tend to exhibit lower volatility.

## 4 Trove Logic

### 4.1 Initial Interest Rate Distribution

In the Liquity V2 system, trove owners have the autonomy to set their own interest rates, constrained by redemption requirements. We define the initial distribution of interest rates as a normal distribution  $r \sim N(\mu, \sigma^2)$ , centered around the initial optimized interest rate such that  $\rho_0 = \mu$ .

## 4.2 Initial LTV Distribution

To model the distribution of Loan-to-Value ratios (LTVs), which represents the ratio of debt to collateral, we employ an approximately normal Beta distribution  $LTV \sim B(\alpha, \beta)$ . The parameters  $\alpha$  and  $\beta$  of the Beta distribution are chosen to match the desired LTV distribution. For approximating a normal distribution, we set  $\alpha = \beta$ , thereby implicitly setting  $\mu \approx 0.5$ , given the confined distribution within the  $[0, 1]$  range, with a variance of  $\frac{1}{4(2\alpha+1)}$ . This distribution is further adjusted to adhere to the maximum LTV constraint allowed within the protocol, typically 91%, thereby shaping user behavior accordingly.

## 4.3 Leverage Demand

$\lambda$  is a normalized value representing the global demand for leverage among participants contributing to the borrowing of stablecoins. This demand is modeled as an Ornstein-Uhlenbeck (OU) process, with parameters fitted based on historical funding rates observed on ETH/USDT Binance Perpetual Futures (Perps).

### Historical Leverage Demand

The historical leverage demand is defined as the normalized equivalent of the 8-hour funding rate on Binance, expressed as:

$$\lambda_t = \frac{Funding\_Rate_t}{\max_{t' \in [0, T]} (Funding\_Rate_{t'})}$$

Here,  $\lambda_t$  represents the normalized funding rate at time  $t$ , scaled by the maximum funding rate observed over the period  $[0, T]$ . This normalization ensures that  $\lambda_t$  ranges between  $[-1, 1]$  and provides a standardized measure of leverage demand.

The funding rate ( $Funding\_Rate_t$ ) reflects the rate at which users are willing to pay to long or short ETH. It captures the cost or reward of maintaining a leveraged position, indicating the market's appetite for leverage over time. The traditional binary nature of the 8-hour funding rate, which is either positive or negative above a minimal threshold, simplifies the representation by focusing on the directional demand for leverage rather than precise deviations from the basis.

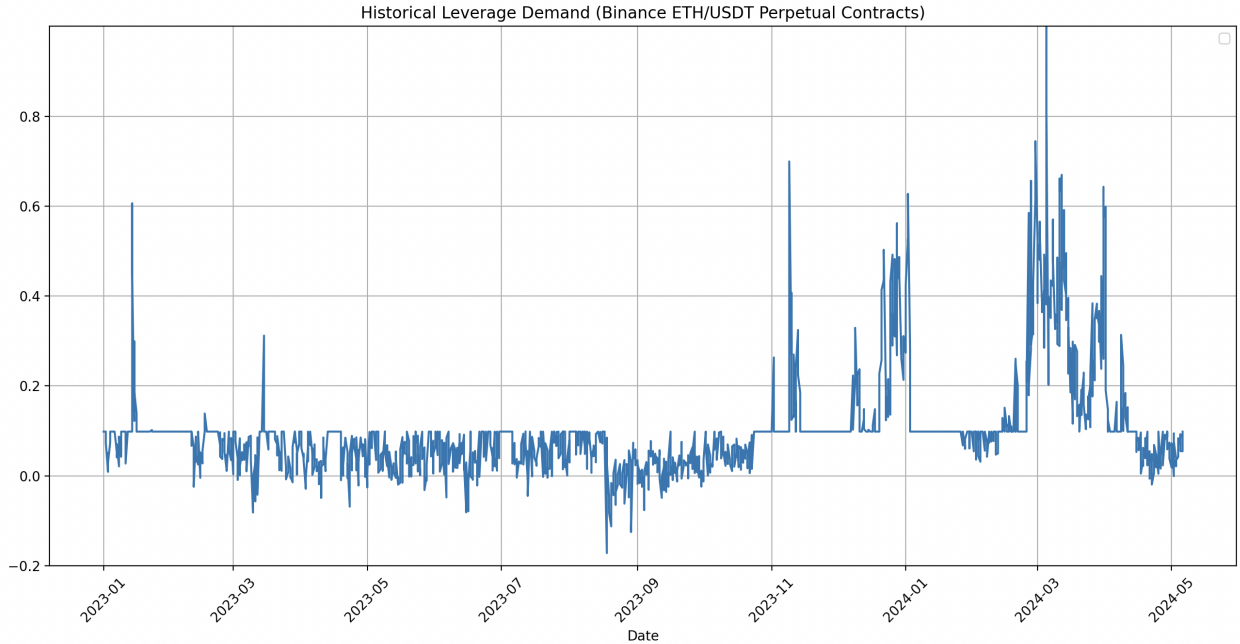


Figure 3: Historical Leverage Demand (Binance ETH/USDT Perpetual Contracts)

#### 4.4 Borrowing Demand Function

We define the infinitesimal change in borrowing demand at a given timestep,  $d\tilde{\lambda}_{i,t}$ , as a function of the global leverage demand,  $\lambda_t$ , and the relative difference between the borrower's interest rate,  $r_{i,t}$ , and the base rate,  $\rho_t$ . This approach is crucial in isolating the effective demand for borrowing on Liquity from the rest of the market. By considering both the interest rate employed by the user relative to the global rate, we can capture the advantage of borrowing on Liquity compared to the global market. Additionally, by accounting for the global demand for leverage as a whole, we obtain a comprehensive view of the borrowing dynamics. The following differential equation expresses the model:

$$d\tilde{\lambda}_{i,t} = \lambda_t \times \frac{\rho_t \times \frac{BOLD}{USD} - r_{i,t}}{r_{i,t}} dt$$

where  $i$  indexes the specific trove, and  $BOLD/USD$  indicates the market price of BOLD. This component is critical as it defines the market value of the obtained capital after swapping minted BOLD on the open market.

To simulate troves adjusting their debt positions over time based on their comfort LTV  $LTV_{c,i}$ , we begin by having all troves open positions at time  $t_0$  at their respective  $LTV_{c,i}$ . Each trove's borrowing demand  $\tilde{\lambda}_{i,t}$  is thus inherently influenced by their respective interest rate  $r_{i,t}$ . This borrowing demand is effectively adjusted over time according to an exponentially decaying function with respect to the trove's LTV, and is bounded by a maximum LTV  $max\_LTV$ .

At each time  $t$ , the LTV of trove  $i$  is updated by adding the value of  $\tilde{\lambda}_{i,t}$  to its LTV at the previous time step  $t - 1$ . Formally, this can be expressed as:

$$\forall LTV_t < max\_LTV, \quad LTV_{i,t} = LTV_{i,t-1} + d\tilde{\lambda}_{i,t}$$

Whereby each trove  $i$  initially starts with  $LTV_{i,t_0} = LTV_{c,i}$ . The borrowing demand is normalized to an infinitesimally small value  $dt$ , which is given by  $\frac{T}{n}$ , where  $T$  is the total time of the simulation and  $n$  is the number of time intervals. This normalized demand is then added to the LTV of each trove at the previous time step, provided that the LTV remains below the maximum threshold  $max\_LTV$ .

In practical terms, this means that the relative increase or decrease in a borrower's LTV will depend on their current LTV. For example, a borrower with an LTV of 0.8 at  $t - 1$  will experience a smaller relative change compared to a borrower with an LTV of 0.5, assuming both face the same interest rates. This is because the effective increase in their borrowing position decreases as their LTV increases, following an exponentially decaying pattern.

By dividing  $\frac{d\tilde{\lambda}}{LTV_{i,t}}$ , we ensure that the relative adjustment in the trove's debt size naturally decays with the size of the position at  $t - 1$ . This reflects a more gradual adjustment for troves with higher initial LTVs, maintaining stability in the system over time.

#### 4.5 Analysis of Scenarios

We consider four distinct scenarios based on the signs of  $\lambda_t$  and the term  $(\rho_t \times \frac{BOLD}{USD} - r_{i,t})$ :

- **Scenario 1:**  $\lambda_t > 0$  and  $\rho_t \times \frac{BOLD}{USD} - r_{i,t} > 0$   
 $d\tilde{\lambda}_{i,t} > 0$

In this scenario, there is positive demand for leverage, and borrowing costs through troves are advantageous compared to alternative lending protocols. This results in an increase in borrowing demand.

- **Scenario 2:**  $\lambda_t > 0$  and  $\rho_t \times \frac{BOLD}{USD} - r_{i,t} < 0$   
 $d\tilde{\lambda}_{i,t} < 0$

Despite positive leverage demand, borrowing costs through troves are higher than the base rate. Users will reduce their borrowing demand and seek more cost-effective leverage strategies.

- **Scenario 3:**  $\lambda_t < 0$  and  $\rho_t \times \frac{BOLD}{USD} - r_{i,t} > 0$   
 $d\tilde{\lambda}_{i,t} < 0$

Here, negative leverage demand indicates market risk aversion or pessimism. Even with advantageous

borrowing costs through troves, users are cautious about taking on additional debt due to potential risks, and they are repaying their debts as a consequence.

- **Scenario 4:**  $\lambda_t < 0$  and  $\rho_t \times \frac{BOLD}{USD} - r_{i,t} < 0$   
 $d\tilde{\lambda}_{i,t} < 0$

Despite negative leverage demand, higher borrowing costs through troves compared to the market compel users to reduce their borrowing demand. The absolute value of the interest rate differential is considered in this scenario.

By examining these scenarios, we gain insight into the nuanced behaviors of users under varying market conditions and their influence on borrowing demand. This approach provides a detailed understanding of the dynamics shaping borrowing behavior in response to changes in leverage demand and interest rate differentials.

## 4.6 Borrowing Demand Scalar

To adequately represent the various behavioral tendencies of users within the protocol, we define a scalar  $\eta \in \mathbb{R}^+$ . This scalar serves as a measure of the level of aggressiveness in user behavior within the system. By adjusting  $\eta$ , we can model and analyze how the system performs under different scenarios, ranging from conservative to aggressive user strategies. This allows us to capture the full spectrum of possible interactions and their impacts on the system's stability and performance.

Plotting the relative change in LTV with varying scalar values, and assuming  $dt = 1$  day:

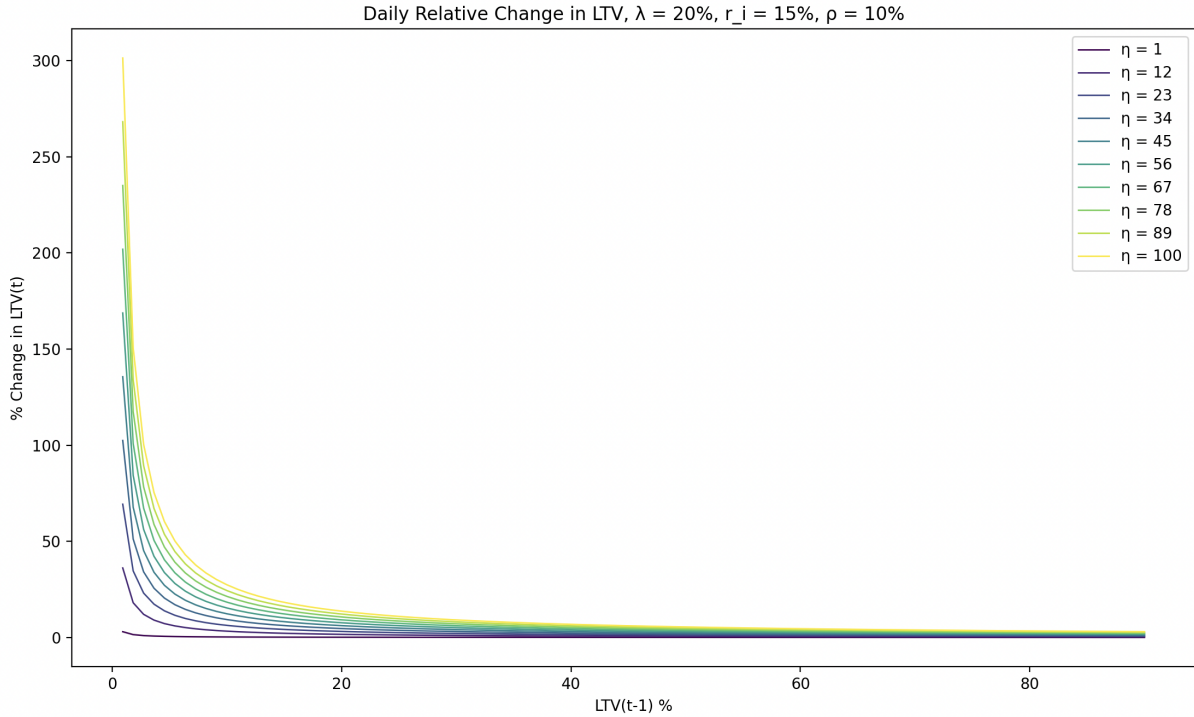


Figure 4: Daily Relative Change in LTV,  $\lambda = 20\%$ ,  $r_i = 15\%$ ,  $\rho = 10\%$

As  $\lambda$  and  $\rho$  scale, while the respective LTV of a user remains on the lower end, the relative change in LTV will scale accordingly, converging at a faster rate as LTV approaches the maximum. The scalar  $\eta$  amplifies this effect, representing the aggressiveness of user behavior.

User behavioral changes based on the difference between  $\rho$  and  $r_i$  are critical. This is characterized by the LTV increasing as a function of the global leverage demand, the absolute deviation between rates, and the overall aggressiveness of users within the Liquity system. The parameter  $\eta$  plays a crucial role in this

context, enhancing the sensitivity of LTV changes to these factors and providing a comprehensive view of the system’s response under various user behaviors.

## 4.7 New Trove Owners

The system initializes all positions at time  $t_0$  based on an initial distribution of Loan-to-Value (LTV) ratios. It adjusts debts according to borrowing demand, modifying the effective LTV based on changes in borrowing demand and ETH price fluctuations. To model collateral inflows over time, new positions are initialized or existing positions are closed at intervals determined by the relationship between  $\rho_t$  and the respective interest rate of a trove owner  $r_i$ , reflecting a percentage change in the total debt within the system. These new positions adhere to distributed comfort levels of LTV and are subject to respective interest rates as a function of targeted debt  $D$ . Mathematically, this can be expressed as:

$$D_t = D_{t-1} \times \left( 1 + \frac{\rho_t \times \frac{BOLD}{USD} - r_{i,t}}{r_{i,t}} \right)$$

This model captures the behavior of new users who mint BOLD to generate yield within the protocol. When the conditions are favorable, characterized by a positive deviation between  $\rho_t$  and the respective interest rates, these users are incentivized to take on new debt to maximize their returns. Conversely, in adverse conditions, where the deviation is negative, these users are likely to revert their course, closing their positions to avoid unfavorable outcomes.

## 4.8 Stability Pool and Base Rate Arbitrage

To realistically model the inflows and outflows to and from the stability pool, given by the yield deviation from the base rate, an aggregate SP agent deposits BOLD if  $SP\_APY > \rho$ , and withdraws funds otherwise. The size of the deposited/withdrawn amount is randomized with an expected value proportional to the relative difference between the yield and the base rate.

Defining a random variable  $\Delta SP(t)$  that represents the net change (deposit or withdrawal) in the Stability Pool at time  $t$ , the decision to deposit or withdraw is modeled as:

$$\Delta SP(t) = \begin{cases} \mathbb{E}[SP\_APY(t) - \rho(t)] \cdot Z & \text{if } SP\_APY(t) > \rho(t) \\ -\mathbb{E}[\rho(t) - SP\_APY(t)] \cdot Z & \text{if } SP\_APY(t) \leq \rho(t) \end{cases}$$

where  $Z$  is a n.i.i.d random variable that captures the stochasticity of the deposit/withdrawal size, following a distribution centered around the expected difference  $\mathbb{E}[SP\_APY(t) - \rho(t)]$ .

Regarding the newly minted BOLD as a function of  $\tilde{\lambda}_t$ , we adopt probabilistic assumptions analogous to those used in the interest rate adjustment process. This approach ensures that each user’s yield optimization coincides precisely with the timing of the interest rate adjustment. Consequently, the aggregate SP agent will allocate the newly generated capital into the SP for all  $r_i < \rho < SP\_APY$  at a given time  $t$ . In other words, for all borrowers who expand their debts, the additional BOLD will be absorbed by the SP as long as  $\rho < SP\_APY$ .

# 5 Parameters and Protocol Logic

## 5.1 Stability Pool Yield Split

In the Liquity V2 design, the system aims to efficiently and scalably incentivize the stability pool by directing a fixed percentage of the aggregate interest into it. This approach helps the stability pool maintain a relative percentage of backing that aligns with the underlying risks of the protocol.

When there is high demand for leverage on the underlying collateral asset, the expected relative aggregate interest rate paid by trove owners will increase due to the implied redemption risk. Consequently, this results in a higher relative and absolute yield for the stability pool. Conversely, during periods of low demand for leverage, trove interest rates will decrease, leading to a smaller relative stability pool size.



The percentage of the total revenue allocated to the stability pool, denoted as  $\%SP$ , is crucial for defining the optimal stability pool size under adverse conditions. This parameter needs to be examined from the perspective of the various agents involved to ensure appropriate incentives during different market scenarios.

## 5.2 Minimum Borrow Rate

The minimum borrow rate  $r_{min}$  is defined at the protocol level as the minimum borrow rate for which a respective trove owner can take on. With the introduction of automated interest rate strategies with respect to redemption risk as a function of the BOLD oracle price, we examine the effect of potential outcomes under various minimum borrow rate values.

## 5.3 Redemption Fee

Much like Liquity V1, in order to create a dynamically bounded game-theoretical mechanism to deter users from taking advantage of the system's market-priced leniencies, such as setting interest rates that are too low, Liquity V2 introduces a similar redemption mechanism that effectively deters the price from going below a specific threshold, subject to the effective redemption fee and the democratized constraint within the system. This works by permissionlessly incentivizing any external actor to maintain the system stability by buying up discounted BOLD and redeeming for user collateral at a \$1 fixed rate, subject to a dynamic redemption fee  $f_t$  at some time of redemption  $j$ .

$$f(t_j) = f_{min} + b(t_j)$$

Where  $f_{min}$  is the minimum redemption fee, and  $b(t_j)$  is a dynamic fee component that scales with respect to the redeemed amount  $m$  relative to the total supply  $n$ , subject to a scaling parameter  $\alpha$ :

$$b(t_j) = b(t_{j-1}) + \alpha \times \frac{m}{n}$$

For instance, assuming  $f_{min} = 0.5\%$  and  $\alpha = 0.5$ , the dynamic redemption fee linearly scales as follows:

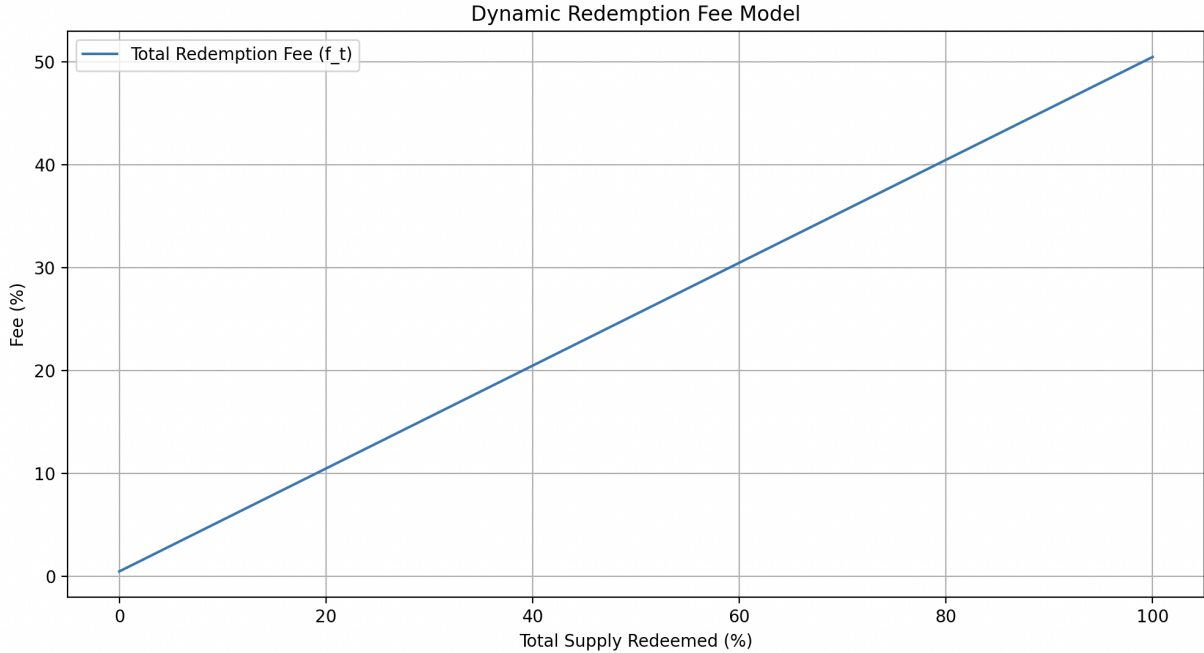


Figure 5: Dynamic Redemption Fee Increase

Thereby calculating the effective collateral  $c$  obtained with respect to the oracle price in USD  $p_c(t)$  as follows:

$$c = m \times p_c(t_j) \times (1 - f(t_j))$$

### Redemption Fee Decay

On the contrary, the dynamic fee component  $b(t)$  decays exponentially over time subject to some decay speed parameter  $\beta$  that dictates how fast the fee decays:

$$b(t_i) = b(t_{i-1}) \times \beta^{\Delta_t}$$

Where  $\Delta_t$  is the time span (modeled as hours) at which the dynamic fee decays over time. In the plot below, we model various  $\beta$  values, subject to some half-life, indicating the speed of fee decay. Note that Liquity V1 employed a  $\beta$  value with a half-life of 12 hours, or approximately 0.944.

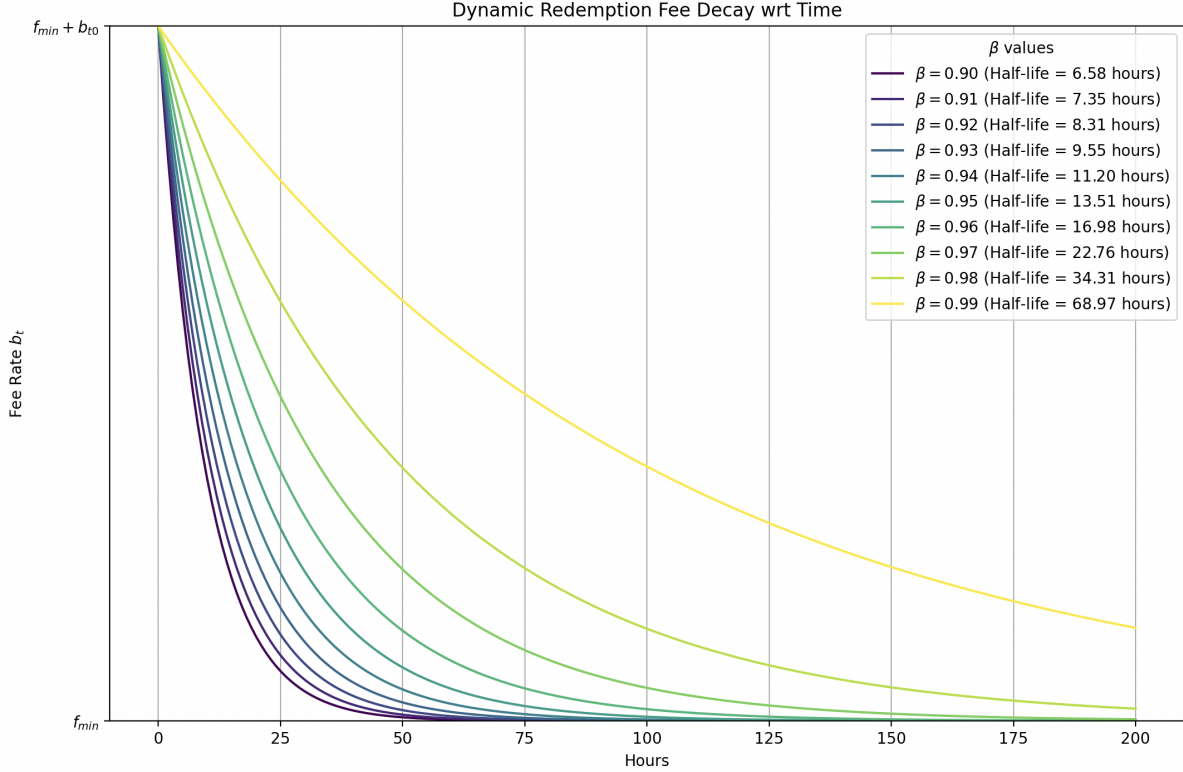


Figure 6: Dynamic Redemption Fee Decay wrt Time for Different  $\beta$  Values

### 5.4 Ordered Set of Troves

Unlike Liquity V1, where redemption queue ordering is defined with respect to the collateral ratio of the user in Last In, First Out fashion (LIFO), Liquity V2 orders troves with respect to the respective interest rate of the user.

Let  $Q = \{\pi_1, \pi_2, \dots, \pi_n\}$  such that  $r(\pi_1) \leq r(\pi_2) \leq \dots \leq r(\pi_n)$ , with each trove  $\pi \in Q$  having an associated interest rate  $r(\pi)$  and debt size  $D(\pi)$ .

For a given redemption amount  $m$ , the subset  $Q_m \subseteq Q$  is redeemed against, such that:

$$S(Q_m) = \sum_{\pi \in Q_m} D(\pi) \leq m$$

where,

$$Q_m = \{\pi \in Q : S(Q_m) \leq m \text{ and } S(Q_m \cup \{\pi\}) > m \text{ for any } \pi \notin Q_m\}$$

ensuring that the selected troves do not exceed the redemption limit  $m$ , and  $Q_m$  is the largest possible subset that can be redeemed up to  $m$  (i.e., partial inclusion of troves that satisfy this constraint).

## 5.5 Target Debt in Front

As a trove owner, the act of being redeemed against effectively fully or partially siphons away the ETH/LST collateral put up in exchange for repaying BOLD liabilities at a 1:1 rate, with some small error with respect to the market price of BOLD that is approximately  $f(t_j) * \frac{D(\pi)}{m}$ , under the assumption that the redeemer acted rationally and redeemed until no longer profitable. However, this partially or completely neutralizes the implied delta exposure of the trove owner, thereby deterring the initial targeted portfolio strategy employed. Thus, given the set of troves is ordered with respect to the interest rate of a respective user, when a trove  $\pi$  is initialized within the simulation, the implementation logic defines a target debt in front coefficient  $\theta \in [0, 1]$  to avoid being redeemed against that is n.i.i.d through an approximately normal beta distribution.

## 5.6 Global Redemption Risk

When the price of BOLD scales downward, thus leading to redemption events, users will naturally attempt to revert to  $\theta$  and thus target some higher interest rate. As a result of the user payoff, we model the global redemption risk  $\gamma \in [0, 1]$  as a normalizing sweep parameter that dictates the probability of reversion to  $\theta$ .  $\gamma$  is thus defined as the relative redemption volume within a 7-day time-frame against a defined maximum EMA redemption volume:

$$\gamma_t = \min \left( \frac{\sum_{i=t-6}^t m_i}{\text{max\_EMA\_redemption\_volume}}, 1 \right)$$

The result for this coefficient is then capped at 1, as can be interpreted above, ensuring that  $\gamma_t$  remains within the range of 0 to 1, where a higher value indicates a higher relative redemption risk.

When taking the relative difference between the current and target debt in front, we implicitly model the elasticity of a respective trove owner  $k$ , subject to his initial targeting games, such that users with lower  $\theta$  values will, in expectation, exhibit more elasticity and thus will aim to update interest rates and minimize their interest accrual accordingly. Unlike V1's ordering dynamics with respect to the LTV of the position, whereby a user only had to be aware of redemption risks and collateral price drawdowns, Liquity V2's ordering can alternatively be modeled through an implicit minimization problem subject to  $\theta$ . In other words, the effective interest rate employed by an updating user to make their interest rate cheaper while adhering to redemption ordering will be  $\epsilon$  greater than the trove owner just below, in order to optimize for both interest rate minimization and risk of redemption minimization:

$$\begin{aligned} & \text{minimize} && r(\pi_k, \theta_k) \\ & \text{subject to} && r(\pi_k, \theta_k) \geq r(\pi_{k-1}) + \epsilon \\ & && \frac{\sum_{\pi_i \in Q, \pi_i < \pi_k} D(\pi_i)}{\sum_{\pi_i \in Q} D(\pi_i)} \geq \theta_k \\ & && p_{adj\uparrow_t} = \left( 1 - \frac{\sum_{\pi_i \in Q, \pi_i < \pi_k} D(\pi_i)}{\sum_{\pi_i \in Q} D(\pi_i)} \right) \times \gamma_t \end{aligned}$$

Where  $p_{adj\uparrow}$  indicates the probability of adjustment and thus a higher relative trove interest rate.

## 5.7 Lowering Interest Rates

Conversely, if the price of BOLD maintains a healthy value within the simulation, thereby inducing minimal redemption volume, while  $\theta$  is less than the current debt in front, then users will naturally attempt to minimize their interest rates back to the target with a probability equivalent to the absolute deviation between the market priced rate  $\rho_t$  and the respective trove interest rate  $r_{k,t}$ , with an additional probability of reducing rate sweep parameter  $\phi$ . Thus, the optimization problem has the following constraints:

$$\begin{aligned}
& \text{minimize} && r(\pi_k, \theta_k) \\
& \text{subject to} && r(\pi_k, \theta_k) \geq r(\pi_{k-1}) + \epsilon \\
& && \frac{\sum_{\pi_i \in Q, \pi_i < \pi_k} D(\pi_i)}{\sum_{\pi_i \in Q} D(\pi_i)} \geq \theta_k \\
& && p_{adj \downarrow t} = (r_k(t) - \rho(t)) \times \phi
\end{aligned}$$

Thus, the probability of adjusting rate is directly correlated with the underlying volatility that  $\rho$  introduces, thereby indicating a more efficient market when  $\rho_t$  is higher.

## 5.8 BOLD Above Peg

A third condition arises in the event that the BOLD price goes above peg, such that the risk of redemption is seen as highly unlikely, and thus troves will no longer adhere to  $\theta$  or any of the conditions or constraints defined earlier, and alternatively decrease their interest rates to the minimum value defined. Formally:

$$\begin{aligned}
& \text{minimize} && r(\pi_k) = r_{min} \\
& \text{subject to} && BOLD/USD > \$1.01 \\
& && p_{adj_t} = \phi
\end{aligned}$$

Given that automated strategies are expected to frequently adjust interest rates according to the optimization logic in expectation, the probability of adjusting the rate becomes significantly higher. This is due to the continuous monitoring and real-time decision-making capabilities of these automated systems, which allow them to react swiftly to changes in market conditions and maintain optimal positions based on their programmed criteria.

## 5.9 Monte-Carlo Runs

For each point in the parameter space, we sample the random variables and processes specified above 1000 times:

- Base rate and leverage demand - CIR and OU processes correlated with a 0.5 coefficient.
- ETH price - GBM process.
- Initial LTV distribution.
- Initial borrowing rate distribution.
- Stop loss/take profit distribution.
- Rate adjustment - binomial with a parameter proportional to the distance from target debt in front.

### Result Representation

- The presented results are the 95th percentile of the relevant metric.
- For minimum BOLD price, and redeemed amount, the results presented were measured over the sample that yielded the 95th percentile redemption volume.
- For bad debt, max BOLD price, minimum TCR, liquidations, and redeemed amount, the results presented were measured over the sample that yielded the 95th percentile bad debt.

## 6 Simulation Results

### 6.1 Sweep Parameter Combinations

In our exploration of this novel protocol, we are immersed in a highly theoretical environment where key parameters delineate the system's overall flexibility and responsiveness. Our approach involves constructing

matrices that encompass a wide array of sweep parameters across diverse market scenarios, as well as isolated stress scenarios that help us further understand the protocol dynamics during adverse instances. This meticulous simulation framework aims to illuminate the anticipated reactions of users within the system. By scrutinizing these simulations, we seek to extract vital metrics that serve as barometers of protocol health. Furthermore, these insights will guide us in formulating precise recommendations for optimizing specific parameter values. This rigorous process not only enhances our understanding of user behavior but also fortifies our ability to foster a resilient and adaptive protocol design.

## 6.2 Key Derived Parameters

We utilize the following derived parameters, at both the protocol level and the agent level, to define various scenarios:

- $\beta$  = decay speed of redemption fee.
- $\eta$  = borrowing demand scalar.
- $\phi$  = probability of reducing rate.
- $\Delta SP$  = Aggregate Stability Pool agent elasticity.
- $r_{min}$  = minimum borrow rate.
- $\%SP$  = stability pool yield split.

## 6.3 Arbitrage Agent Elasticity and Speed of Redemption Fee Decay

Parameters Swept  $\Delta SP$  and  $\beta$

### Rationale:

The elasticity of the stability pool yield relative to the base rate, combined with varying redemption decay speeds, significantly impacts the system's behavior. Increased elasticity allows the system to more effectively minimize the deviation between the global interest rate and the stability pool yield. This heightened efficiency can lead to instances of large redemptions, particularly during significant spikes in base rates. These large redemptions increase outflows, potentially causing downward price movements. Understanding and managing this elasticity is crucial for maintaining system stability and mitigating the risks associated with rapid interest rate changes, as well as addressing the impact of liquidations on stability pool contraction.

In this context, the speed of redemption decay becomes imperative in maintaining the peg. Faster decay speeds can help quickly absorb the selling pressure and stabilize the system by reducing the amount of BOLD in circulation more rapidly. Conversely, slower decay speeds may prolong the period of instability, as the system takes longer to adjust to the increased selling pressure and restore equilibrium. **Interesting**

### Metrics

- `total_redeemed_amount` - represents the relative redemption volume concerning the total BOLD supply at initialization. The values in the cells indicate the proportion of BOLD that has been redeemed relative to the initial total supply of BOLD.
- `min_BOLD_Price`.

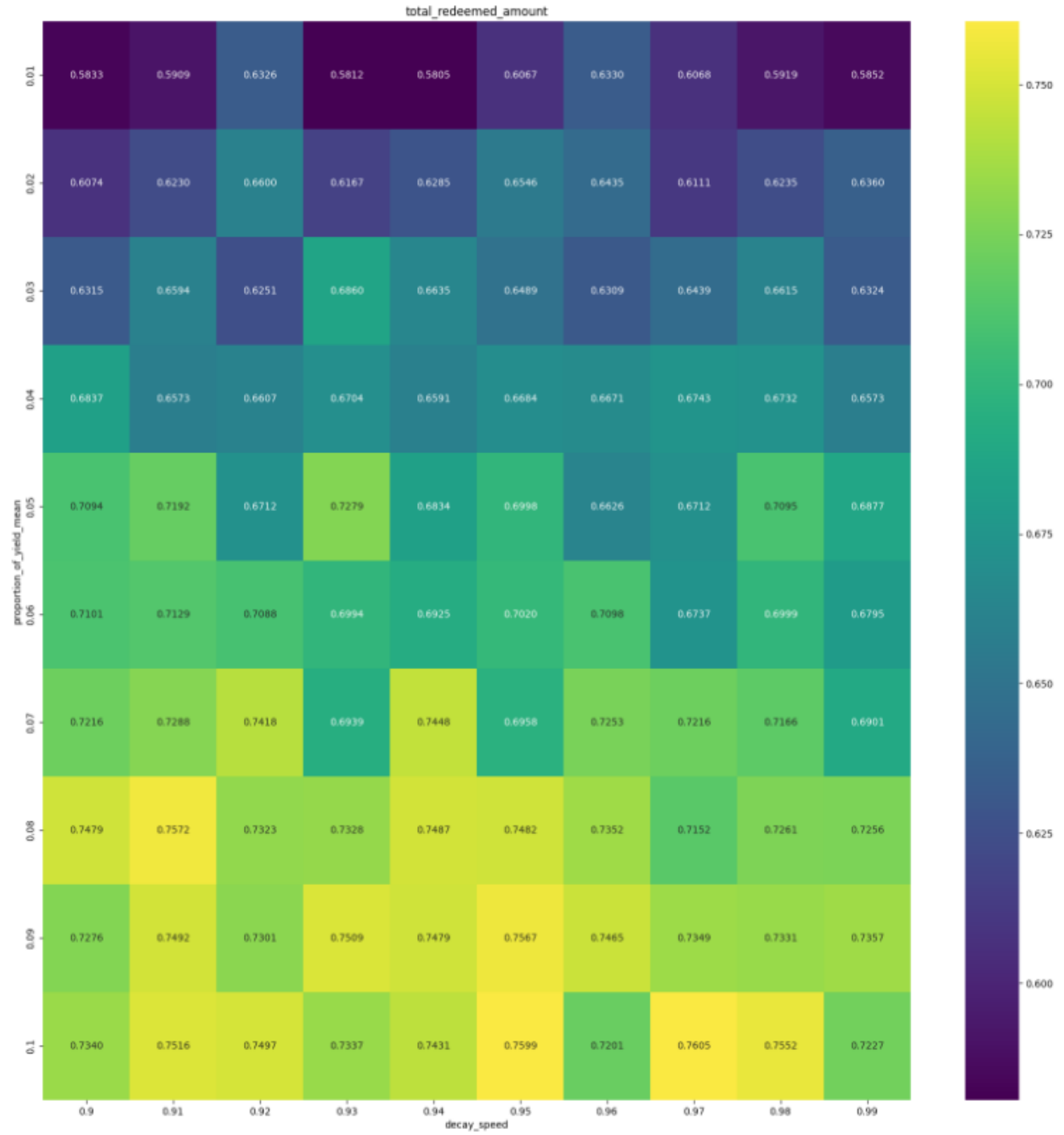


Figure 7: Heatmap of total redeemed amount for various combinations of decay speed and aggregate SP agent elasticity.

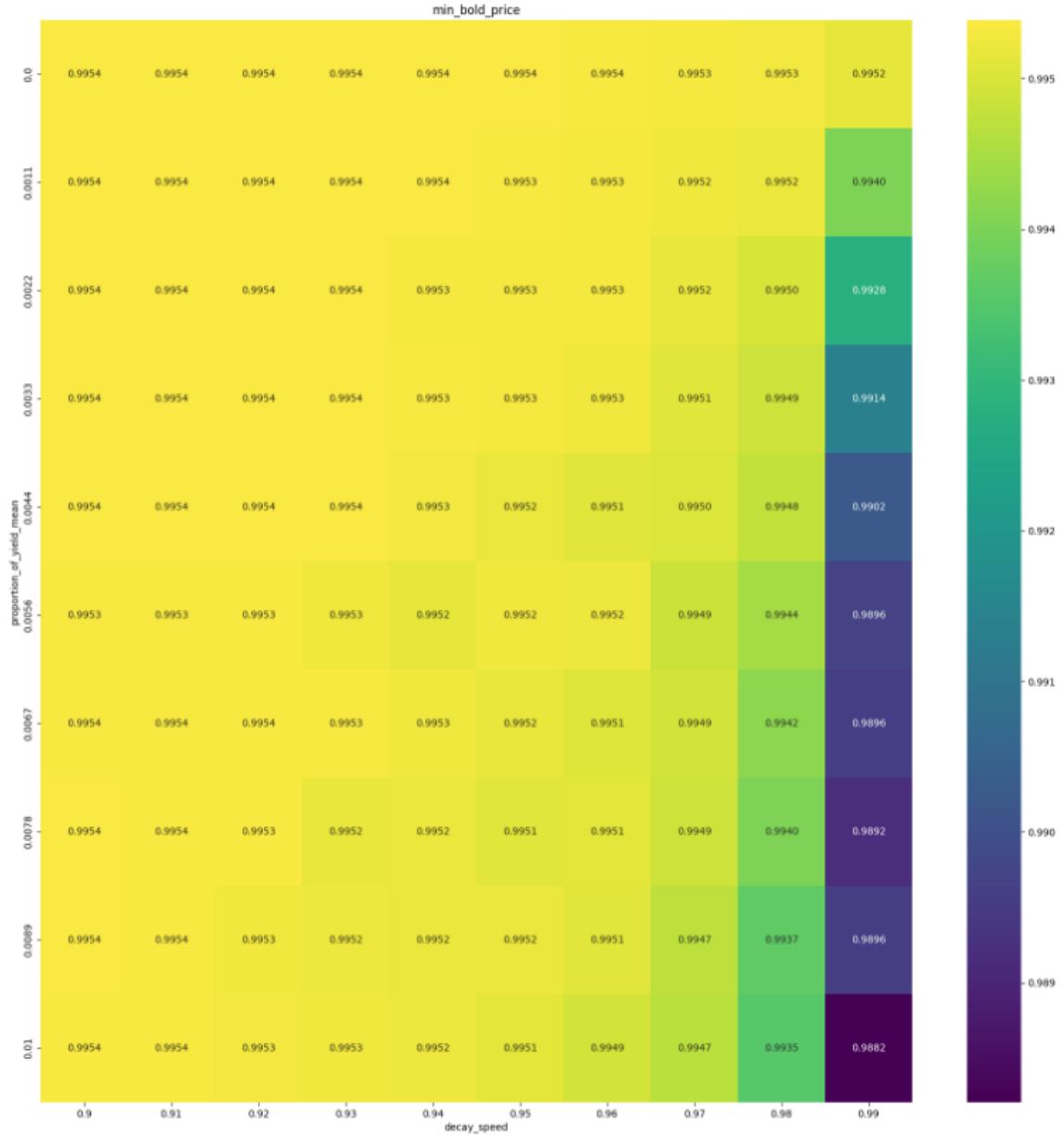


Figure 8: Heatmap of minimum BOLD price for various combinations of decay speed and aggregate SP agent elasticity.

## 6.4 Aggressiveness in Borrowing Demand and Speed of Redemption Fee Decay

**Parameters Swept:**  $\eta$  and  $\beta$

**Rationale:** The aggressiveness in user borrowing demand, coupled with varying redemption decay speeds, plays a significant role in the system's dynamics. When users aggressively amplify their Loan-to-Value (LTV) ratios and mint new debt, it can result in heightened selling pressure on BOLD, as well as increased liquidation events. This increased selling pressure can trigger redemption events, where users redeem BOLD for the underlying collateral to restore the desired collateralization levels.

**Metrics:**

- Min BOLD Price

- total\_redeemed\_amount
- bad\_debt
- total\_liquidated\_amount

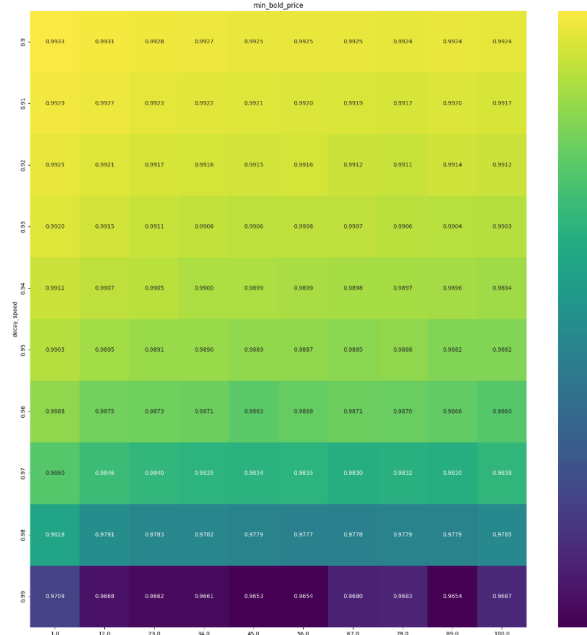


Figure 9: Min BOLD Price Heatmap

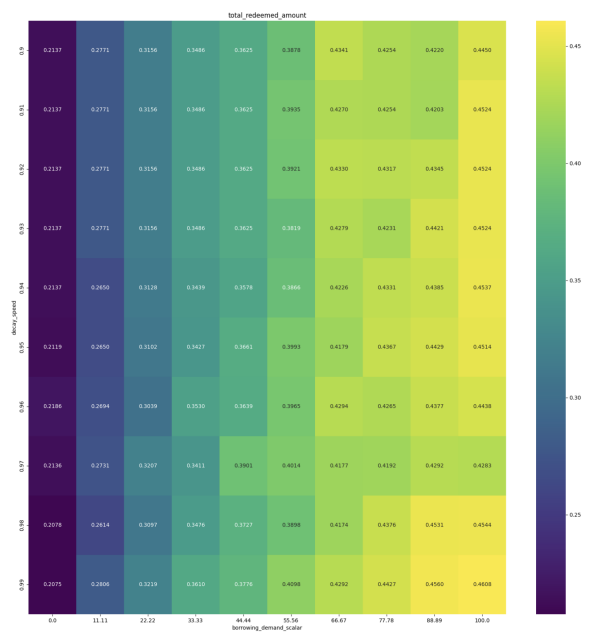


Figure 10: Total Redeemed Amount Heatmap

## Impact of Minimum Borrow Rates

Below, we present two sets of heatmaps: on the left-hand side, the system operates under a minimum borrow rate of 0%, while on the right-hand side, it operates under a minimum borrow rate of 0.5%. These visualizations illustrate the impact of different minimum borrow rates on the system's performance.

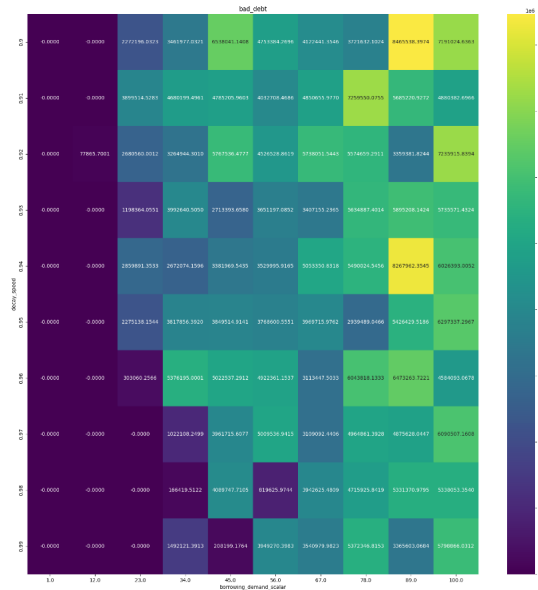


Figure 11: Bad Debt with a Minimum Borrow Rate of 0%

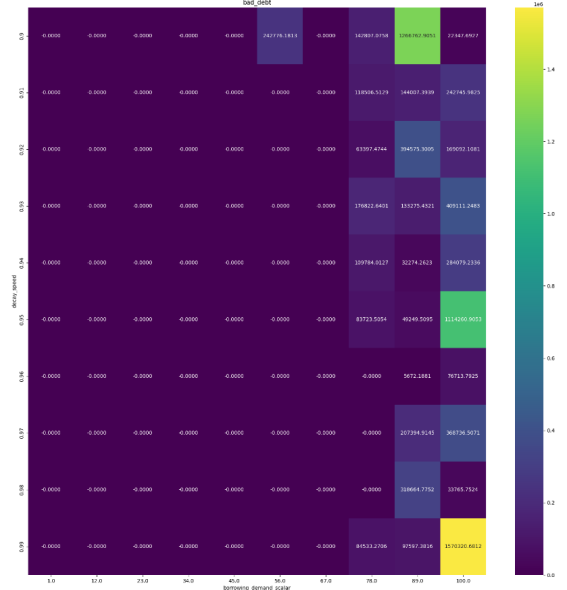


Figure 12: Bad Debt with a Minimum Borrow Rate of 0.5%



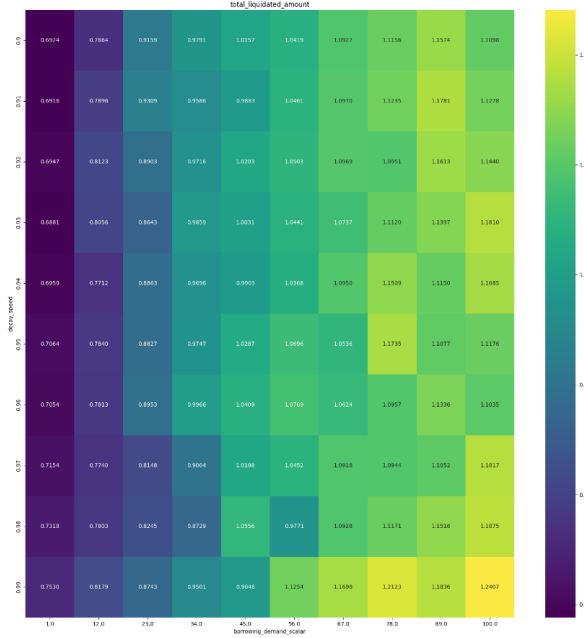


Figure 13: Total Liquidated Amount with a Minimum Borrow Rate of 0%

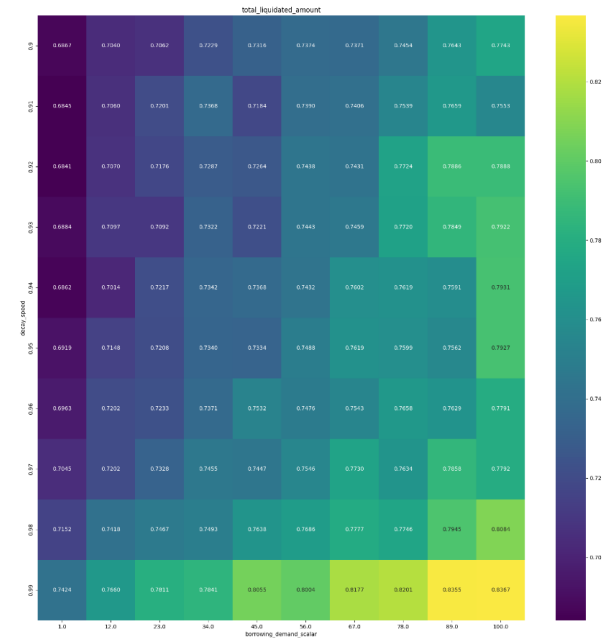


Figure 14: Total Liquidated Amount with a Minimum Borrow Rate of 0.5%

Importantly, the heatmaps reveal a significant decrease in the accrual of bad debt in the worst-case scenarios when the minimum borrow rate is set at 0.5%. This observation suggests that a higher minimum borrow rate enhances the system’s resilience during adverse market conditions.

## 6.5 Elasticity of Stability Pool Coupled with Elasticity of Trove Owner Interest Rate Adjustments

**Parameters Swept:**  $\Delta SP$  and  $\phi$

**Rationale:** Trove owner elasticity refers to the likelihood that trove owners will reduce their rates in response to changes in the base rate at any given step. This elasticity indicates how sensitive trove owners are to fluctuations in the base rate, influencing their decisions to adjust borrowing rates accordingly. On the other hand, aggregate SP agent elasticity pertains to the probabilistic reaction of the stability pool agent to deviations between the base rate and the stability pool yield. This elasticity measures how the stability pool agent responds to discrepancies in yield, adjusting their actions to maintain equilibrium within the system. Together, these elasticities play an important role in determining the overall responsiveness and stability of the system in reaction to varying market conditions.

**Metrics:**

- Max BOLD Price
- Stability Pool Size
- Redemption volume

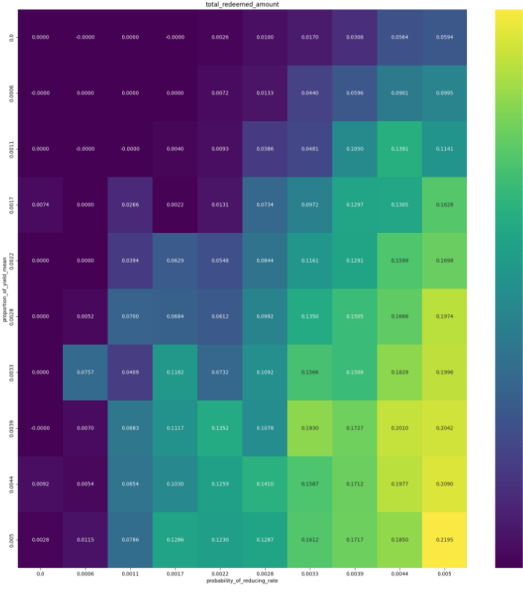


Figure 15: Total Redeemed Amount

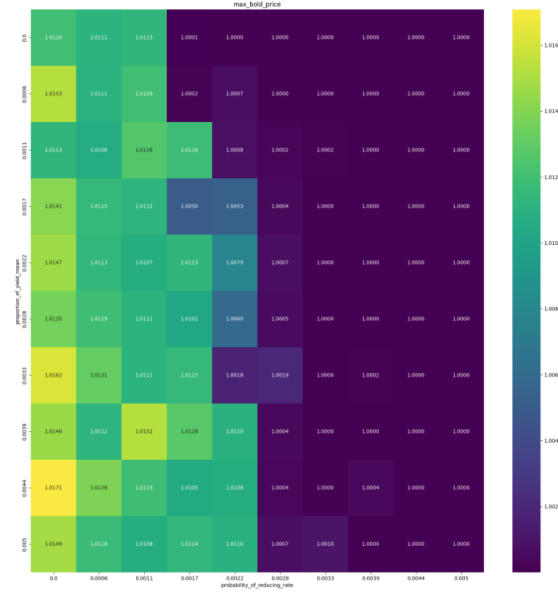


Figure 16: Max BOLD Price

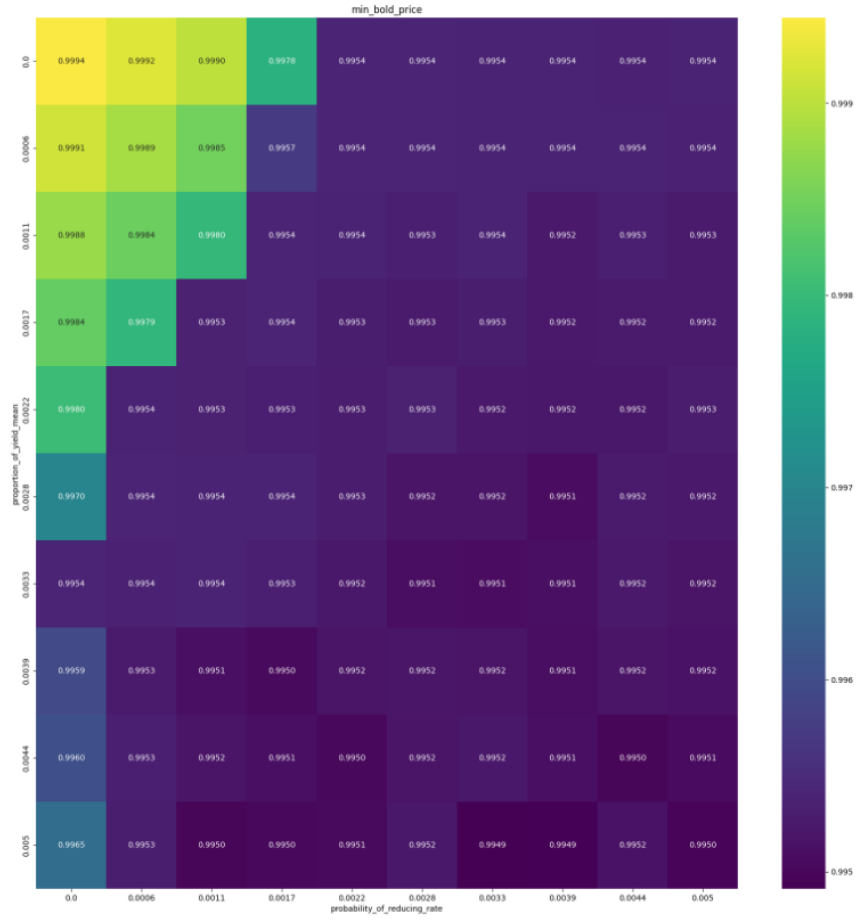


Figure 17: Min Bold Price

## 6.6 Protocol Incentives and Limitations

**Parameters Swept:**  $\%SP$  and  $r_{min}$

**Rationale:**

Gauge the optimal level of incentives given to the SP coupled with the minimum borrowing rate defined for trove owners, such that the system is adequately incentivized and maintains sufficient backing in the worst-case scenario.

**Metrics:**

- Min BOLD Price
- Stability Pool Size
- Redemption volume

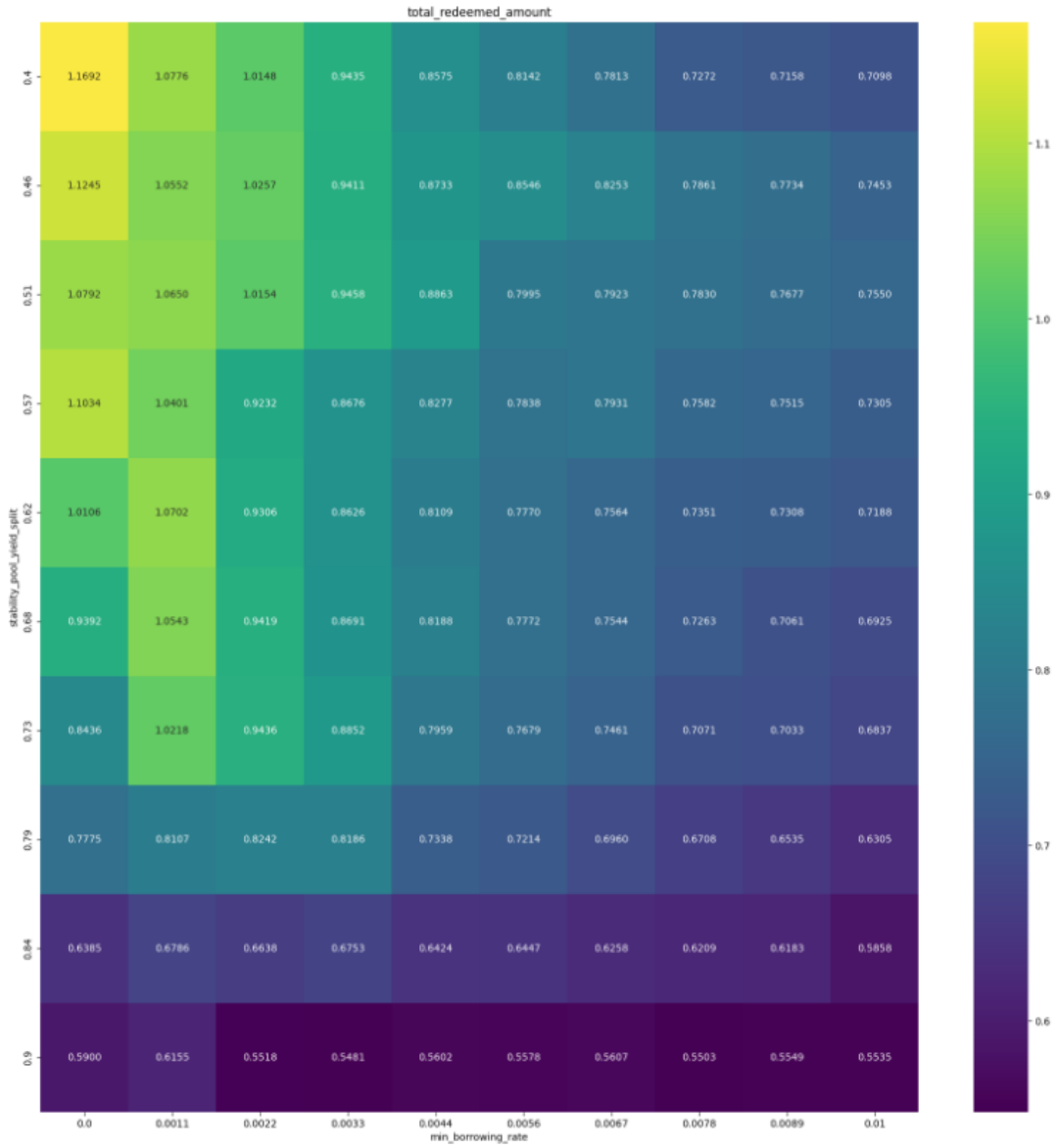


Figure 18: Total Redeemed Amount

Intuitively, one could define the minimum borrow rate as an absolute revenue threshold for the protocol. Concurrently, the stability pool yield split serves as an explicit incentive mechanism to supply idle capital to the stability pool, ensuring it maintains an adequate percentage of the total BOLD supply. Consequently, when there is less gross revenue going to the protocol and a smaller percentage of revenue accruing to the stability pool, it inherently leads to an increase in redemption events. With a reduced internal incentive to hold onto BOLD, due to limited yield sources, and the minimum borrowing rate converging to lower levels when the BOLD price scales above a certain threshold, more redemption events are triggered. This dynamic underscores the importance of balancing revenue distribution and incentive mechanisms to maintain system stability and minimize excessive redemptions.

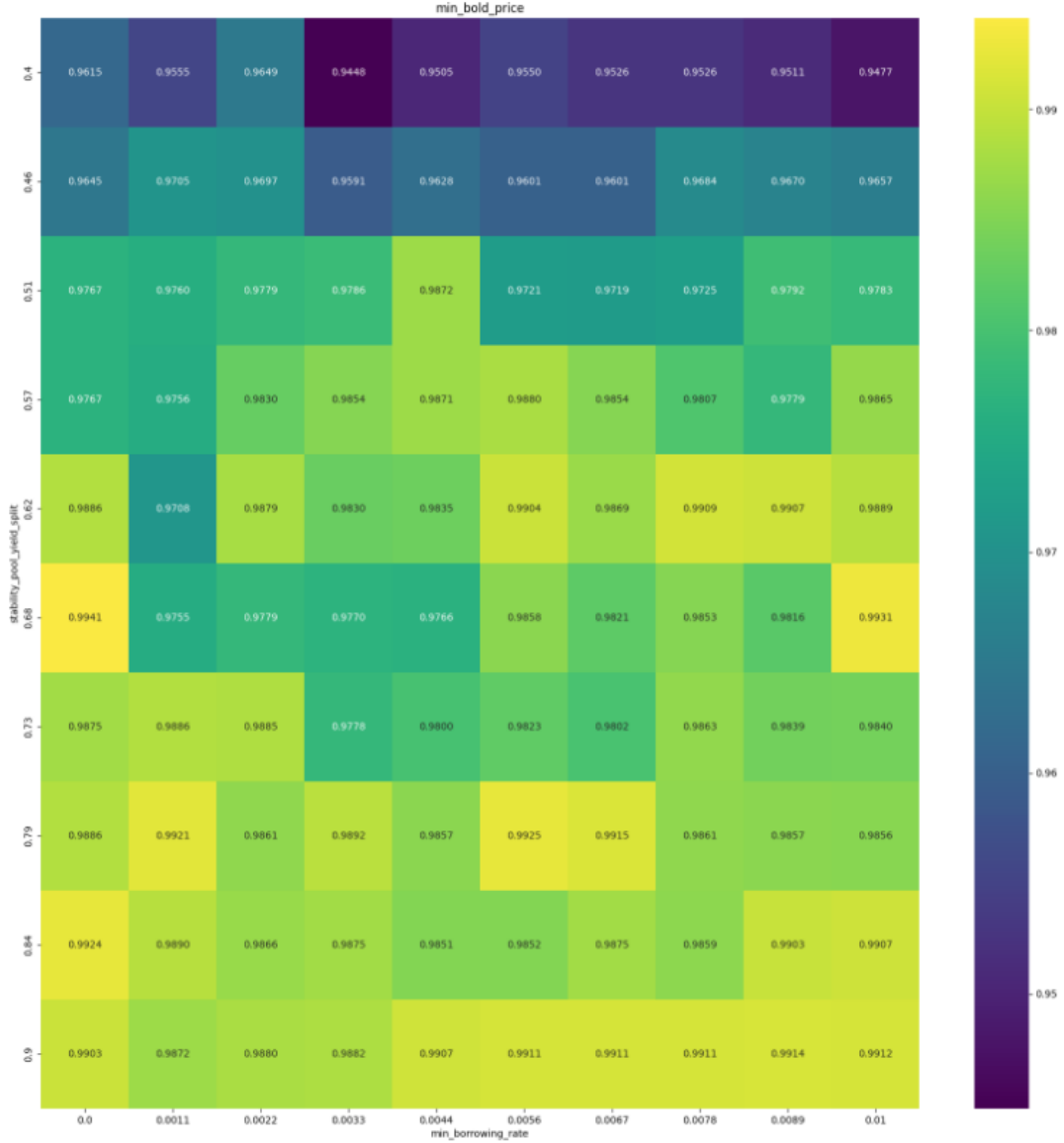


Figure 19: Min BOLD Price

The BOLD price, while not following as much of a gradient, naturally provides a relationship between its value and the total redeemed amount. Similar to the total redeemed amount, as the stability pool yield split gets smaller, the incentive to hold minted BOLD instead of selling to take advantage of leverage elsewhere,

as well as buy on the open market for non-redemption purposes, decreases.

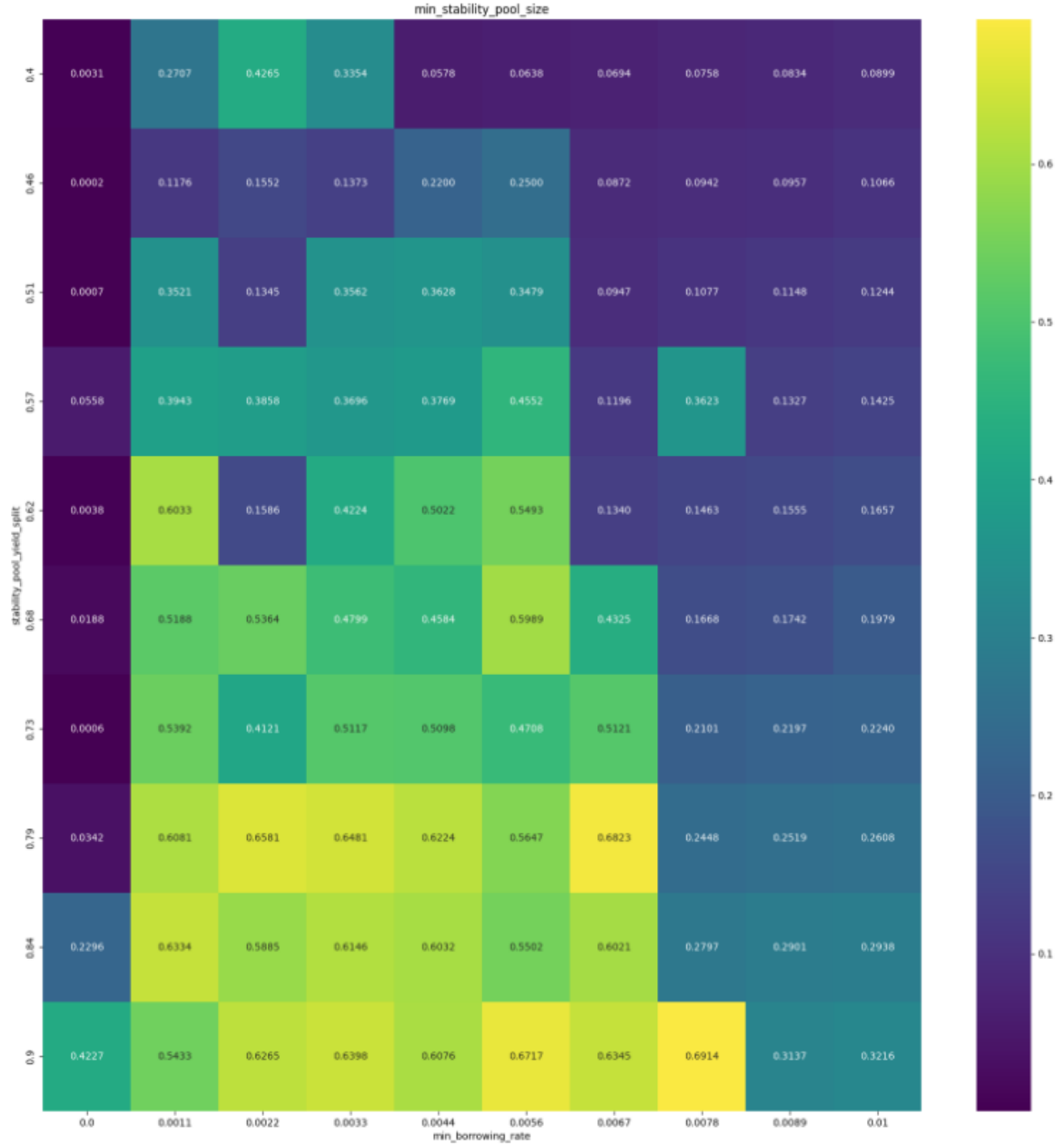


Figure 20: Min Stability Pool Size

In terms of the minimum stability pool size, however, the pattern is quite different. While a low minimum borrow rate coupled with a low stability pool yield split makes it such that the total capital in the stability pool converges to a low value, one would think that the larger the absolute revenue (i.e., minimum borrow rate) coupled with a high stability pool yield split would lead to a larger stability pool as a result. However, according to our results, a minimum borrow rate that is too high will lead to adverse outcomes relative to a lower minimum borrow rate, as the relative incentive to mint new capital when leverage demand is high thus diminishes comparatively.

## 6.7 Modeling Additional Trove Operations

To further optimize the value for minimum borrowing rates, we have added these critical trove operations, which are more reflective of realistic users:

- **Stop loss** - acceptable percentage change from the opening price. Troves stop loss distribution is sampled out of a normal distribution. We have attempted to use historical V1 data to estimate the parameters of the distribution.
- **Adjusting LTV** - repaying loan to return to comfort LTV. Adjusting LTV each time step  $t$  is binomial with a probability:

$$p = \phi * \left| \frac{LTV_{i,t} - LTV_{c,i}}{LTV_{c,i}} \right|$$

Where trove responsiveness is the same parameter  $\phi$  used when adjusting the interest rate.

Unsurprisingly, introducing these actions has mitigated bad debt significantly (0 for any reasonable selection of stop losses and responsiveness parameters). However, in order to get a sense of the effect of minimum borrowing rate at a working point, we have used a “stressed” stop loss distribution with mean = 0.6, std = 0.05, and a synthetic ETH price path of an 80% drawdown over 2 weeks, as shown in the charts below:

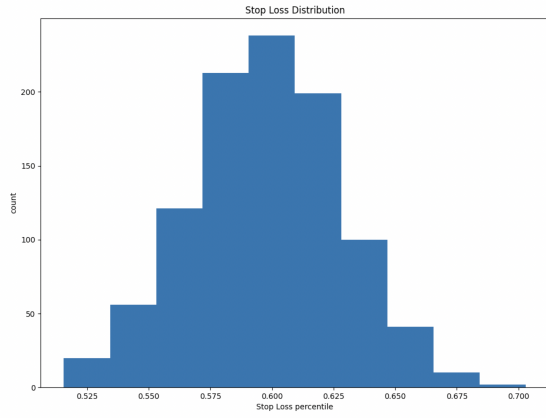


Figure 21: Stop Loss Distribution

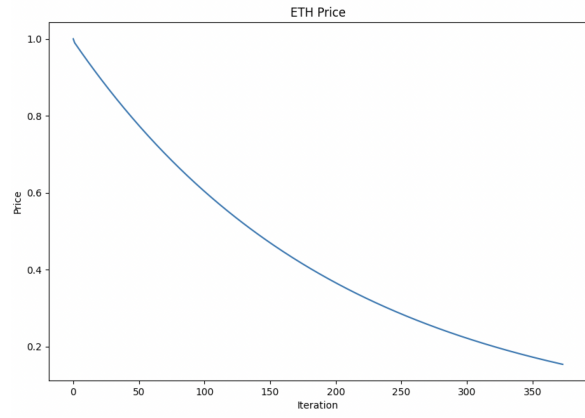


Figure 22: ETH Price

Results

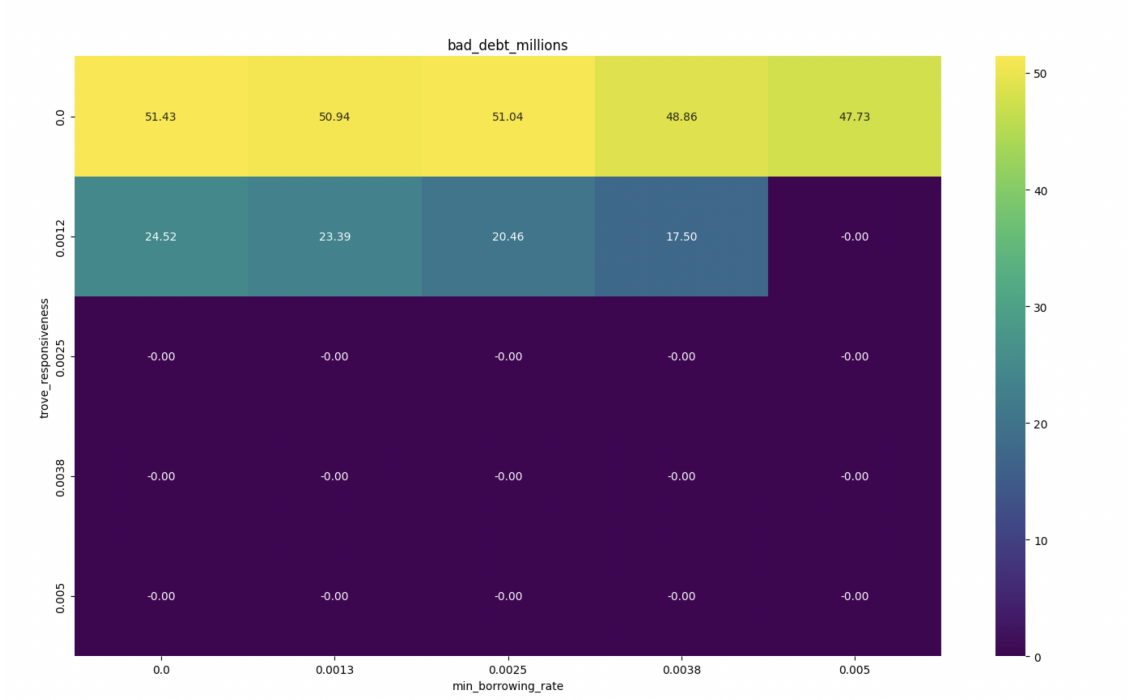


Figure 23: Bad Debt in Millions

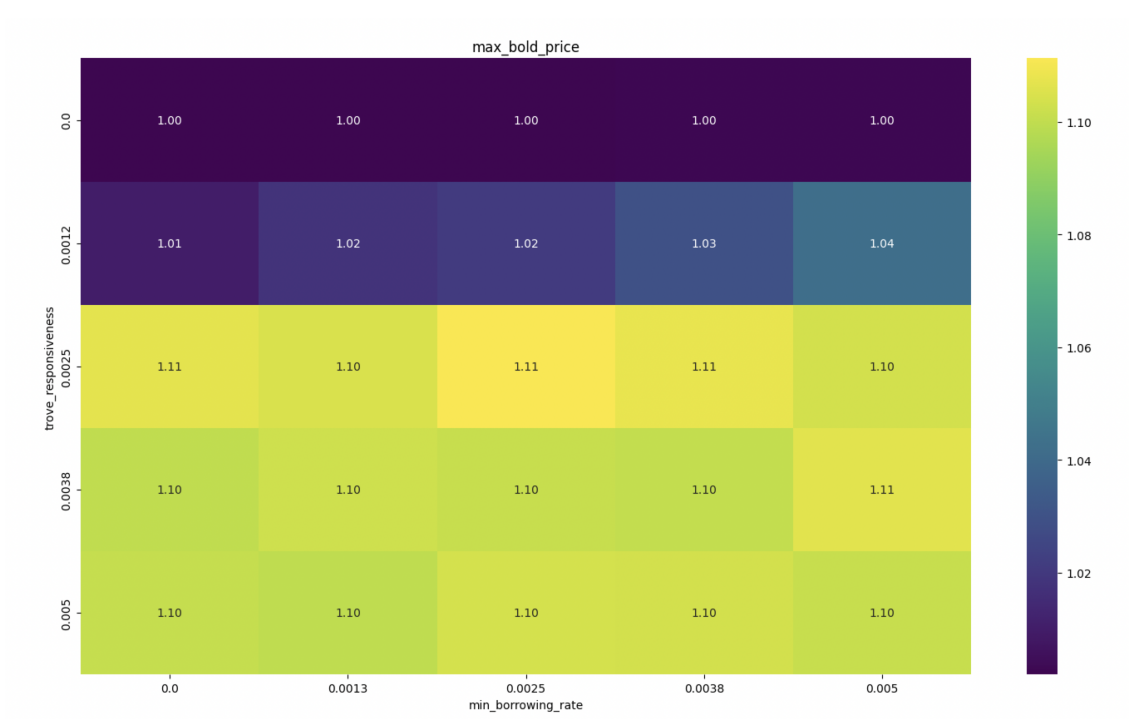


Figure 24: Max BOLD Price

While it's difficult to model users' responsiveness under high volatility regimes and translate the parameter to the actual probability of a trove to adjust LTV and avoid liquidation, it is clear based on historical data that we should expect significant repayments and lower bad debt than in previous simulations. We also see the effect of the repayments on BOLD price, with increasing demand.

## 7 Results and Insights

### 7.1 Implementing a Minimum Interest Rate

We mathematically analyze the dynamics observed among the agents to gain insights into defining a minimum interest rate. This analysis stems from the continuously incentivized stability pool yield, which results in high variance in stress scenarios, and the strategic interest rate adjustments that can lead to detrimental outcomes within the protocol. By understanding these dynamics, we can better define and implement a minimum interest rate that balances incentives and stability, mitigating the risks associated with high variance in stability pool yields and the implications of strategic interest rate setting.

#### BOLD Upward Price Pressure and Stability Pool Dynamics

The stability pool APY jumps stemming from the percentage of cumulative interest rates when the stability pool contracts due to liquidation events. Mathematically speaking,

$$\forall t \in T, SP_{t-1} > SP_t \mid Liquidation_t \rightarrow APY_t > APY_{t-1}$$

Here,  $SP_t$  represents the size of the stability pool at time  $t$ . During periods of significant liquidations, the deviation between  $SP_{t-1}$  and  $SP_t$ , or  $\Delta SP_t$ , increases, causing  $\Delta APY_t$  to rise as well. This relationship is crucial as it illustrates how the protocol incentivizes stability and risk mitigation during market stress, aligning the evolving interests of users who provide liquidity to the stability pool.

#### Impact of Expected Stability Pool Size and Yield Split on APY Variance

The size of the stability pool  $SP_t$  directly impacts the magnitude of APY changes. Smaller pools exhibit more significant percentage changes in APY compared to larger pools, given the same absolute change in liquidated assets. Formally, the expected deviation in APY  $\Delta APY_t$  as a function of the yield split  $\%SP$  can be expressed as:

$$\Delta APY_t = APY_t - APY_{t-1} \propto f(\%SP) \times \frac{\Delta SP_t}{SP_{t-1}}$$

Where  $f(\%SP)$  is a monotonically decreasing function of the yield split, and the cumulative annualized gross revenue within the protocol is defined as  $\sum_i^n r(\pi_i) \times D(\pi_i)$ . Under the assumption that  $APY_{t-1}(\%SP_1) = APY_{t-1}(\%SP_2)$ , indicating converged stability pool sizes varying with  $\%SP$ , alongside constant relative liquidation sizes and redeemed trove interest rates, the relative expected deviation between APY values thus decreases monotonically with the yield split  $\%SP$ :

$$\forall \%SP_1, \%SP_2 \in [0, 1], \%SP_1 \leq \%SP_2 \Rightarrow \frac{\Delta APY_t}{APY_{t-1}}(\%SP_1) \geq \frac{\Delta APY_t}{APY_{t-1}}(\%SP_2)$$

Naturally, for smaller stability pools, the absolute change  $\Delta SP_t$  represents a larger percentage change. Formally, if the expected stability pool sizes  $\mathbb{E}[SP_{t-1}(\%SP_1)]$  and  $\mathbb{E}[SP_{t-1}(\%SP_2)]$  are such that  $\mathbb{E}[SP_{t-1}(\%SP_1)] < \mathbb{E}[SP_{t-1}(\%SP_2)]$ , given by  $\%SP$ , we have:

$$\frac{\Delta SP_t}{\mathbb{E}[SP_{t-1}(\%SP_1)]} > \frac{\Delta SP_t}{\mathbb{E}[SP_{t-1}(\%SP_2)]}$$

Therefore, the relative change in the stability pool size is more significant for smaller pools in expectation. This implies that for smaller pools, the impact of  $\Delta SP_t$  is amplified, leading to greater fluctuations in APY. Given that the expected stability pool size  $\mathbb{E}[SP_{t-1}]$  is a function of the yield split  $\alpha$ , the impact of liquidation events on the relative APY and the relative size of the stability pool is dual-faceted:



1. The relative change in APY decreases with increasing  $\%SP$ .
2. The relative size of the stability pool decreases more significantly for smaller pools, which is the case when  $\%SP$  is smaller due to less continuous incentives.

Thus, during rapid collateral price declines where  $SP \rightarrow 0$ ,  $APY \rightarrow \infty$ , smaller stability pool sizes converge faster and exhibit higher yield variance compared to larger pools in expectation. This dynamic is compounded by trove owners rapidly repaying debt during such downturns, influencing overall system stability and asset pricing.

### Comparing Liquity V1 Borrow Fee Dynamics to Liquity V2 Interest Rates

Moreover, comparing borrowing dynamics between Liquity V1 and Liquity V2 reveals contrasting approaches to managing borrower incentives and system stability. In Liquity V1, the upfront borrow fee of 0.5% acts as an effective deterrent against short-term speculative borrowing. This fee structure ensures that borrowers consider the economic viability of their positions relative to market volatility and potential liquidation risks. The fee structure implicitly defines an extrapolated Borrow APY based on the expected duration of trove positions, thereby influencing borrowing behaviors during volatile market conditions, as seen below:

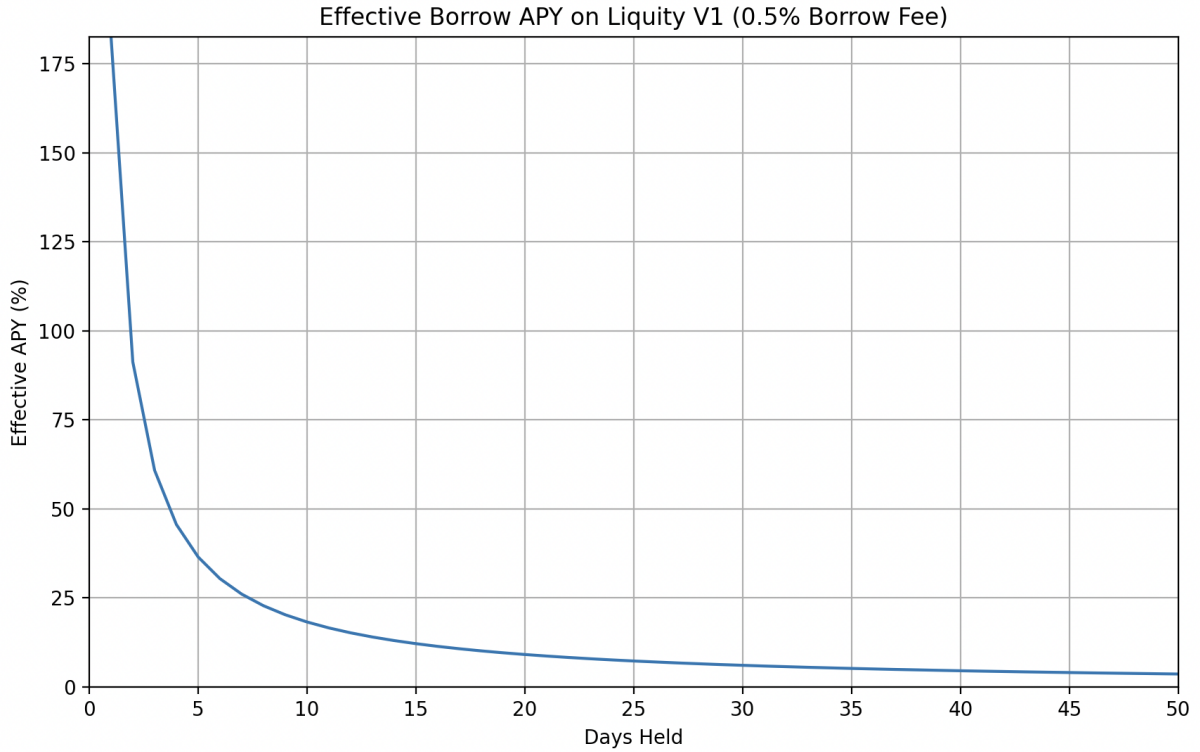


Figure 25: Effective Borrow APY on Liquity V1

Conversely, Liquity V2 trove owners can adjust interest rates dynamically and autonomously, which can impact system health during stress scenarios. Managing risk-adjusted revenue and preventing rent-seeking behaviors during market downturns are critical, necessitating a balanced approach to borrowing fees to maintain system stability. Our borrowing demand mechanism and the initialization of new troves assume that highly elastic users can take on free debt, thereby potentially negatively affecting the system's risk-adjusted health during a market downturn due to minimal proportional interest contributions to the dwindling stability pool.

In a stress scenario where liquidations and implicit debt repayments occur continuously, it is assumed that the automated interest strategy for a large subset of users will adjust automatically if the BOLD price exceeds a specific threshold. Consequently, the aggregate risk-adjusted revenue for new deposits inhibits the stability

pool’s rebounding effect and can lead to additional outflows and liquidation events, resulting in bad debt accrual. As demonstrated in stress test results and Monte Carlo simulations, to mitigate the potential for rent-seeking behavior during significant market downturns, we recommend instituting a minimum interest rate of 0.5%. This fee should strike a balance between not overly taxing the incentive to mint new capital when leverage demand is high while simultaneously protecting against adverse outcomes.

## 7.2 BOLD Downward Price Pressure and Redemption Risks: Less Aggressive Redemption Fees

Our simulation results indicate that decreasing the dynamic fee constraint  $\beta$  resulted in no discernible increase in redemption volumes and thus a minimal contraction in the BOLD supply. Despite these relatively minor changes, the fee adjustment had a substantial impact on improving peg stability, particularly on the downside. This enhanced stability prompted users to adapt their behavior in ways that are markedly different from the patterns observed in V1.

### V1 Dynamics

In V1, avoiding redemption explicitly increased short-term peg stability because trove owners would initiate an LUSD buy on the open market. In a similar simulation environment for V2, the objective function can be defined as follows:

$$\begin{aligned} & \text{minimize} && CR(\pi_k, \theta_k) \\ & \text{subject to} && CR(\pi_k, \theta_k) \geq CR(\pi_{k-1}) + \epsilon \\ & && \frac{\sum_{\pi_i \in Q, \pi_i < \pi_k} D(\pi_i)}{\sum_{\pi_i \in Q} D(\pi_i)} \geq \theta_k \\ & && p_{adj\uparrow_t} = \left( 1 - \frac{\sum_{\pi_i \in Q, \pi_i < \pi_k} D(\pi_i)}{\sum_{\pi_i \in Q} D(\pi_i)} \right) \times \gamma_t \end{aligned}$$

In this context, rate minimization is replaced with LTV minimization. Trove owners buy and repay LUSD on the open market until  $\theta$  is satisfied, allowing the LUSD price to rebound independently. The precise amount of LUSD to be bought up can be mathematically represented through the following equation:

$$\Delta LUSD = D(\pi_k, t-1) \times \left( 1 - \frac{CR(\pi_k, t-1)}{CR(\pi_k, t)} \right)$$

Thereby explicitly contributing to peg reversion.

### Constructing a Middle Ground Between BOLD and Traditional CDP Stablecoins

In contrast, traditional decentralized stablecoins like Aave’s GHO and MakerDAO’s DAI employ a ‘soft peg’ mechanism. They incentivize trove owners to repay debt at a cheaper rate through market purchases in the event of a depeg and periodically adjust interest rate parameters to maintain a \$1 equilibrium. This approach aligns with market-priced interest rates to inhibit extractive behavior that can negatively impact peg health.

Liquity’s BOLD introduces a democratized, autonomous, and algorithmic approach, characterized by very low interest rates and dynamic fee peg constraints based on relative redemption volumes over specific periods. Consequently, the minimum BOLD price is highly sensitive to the decay speed of these dynamic fees; less restrictive fee dynamics lead to lower prices, influenced by user-driven adjustments within the system due to the underlying autonomy.

Our simulations indicate that the overall redemption volume within the system typically remains unaffected by the decay speed. This is because risk-taking trove owners do not actively purchase BOLD on the open market, unlike in V1. Instead, this necessitates users to fine-tune the aggressiveness of their interest rate strategies to manage redemption risks, which are expected to oscillate around the market-priced interest rate.

Therefore, we recommend implementing a lower  $\beta$  value than V1’s 0.944 and adopting a more conservative dynamic fee approach. This strategy aims to mitigate extractive behaviors within the protocol and enhance

peg stability while simultaneously managing typical debt incentives, liquidation risks, and debt payoffs (incentives/haircuts) associated with CDP stablecoins.

## 7.3 Modeling Different LSTs

### Stability Pool Implications

The passive nature of the continuously incentivized stability pool in internalizing liquidation events significantly reduces reliance on on-chain liquidity, thereby mitigating negative externalities that occur within a short timeframe. Our extreme stress tests demonstrate that a stability pool capable of internalizing a substantial amount of debt effectively buffers the system against sudden shocks and market volatility. This includes mitigating the impact of sudden ETH price drops within a very short time window, as well as adverse decorrelation events, such as slashing, that occur within the protocol and the dynamics surrounding these events.

### User Behavior and External Pricing

User behavior is expected to remain consistent whether using a Liquid Staking Token (LST) or ETH, as the underlying mechanisms and incentives remain largely unchanged. By overlaying highly correlated price trajectories within simulations, we can project similar probabilistic outcomes for both assets. Following the Shanghai upgrade, the pricing of LSTs has converged closely with ETH, driven by a fundamental change in market confidence and thus improved liquidity.

This convergence has been reflected in lending protocols such as Aave, which have already begun pricing LSTs 1:1 with ETH in their oracles. Furthermore, these protocols have introduced a generalized "e-mode" for a subset of LSTs with extremely high theoretical Loan-to-Value (LTV) ratios, operating under the assumption that the underlying debt asset is correlated (such as ETH). This adoption of "e-mode" demonstrates a recognition of the convergence between LST and ETH prices and highlights the enhanced liquidity and market confidence in LSTs post-Shanghai upgrade.

The inherent concentration of ETH/LST liquidity on decentralized exchanges (DEXes) facilitates a precise definition of the relative LST collateral through the redemption process. This mechanism ensures that arbitrageurs can effectively redeem LSTs when necessary, as the liquidity structure supports seamless exchanges. Furthermore, the inverse correlation between liquidations and redemption events provides additional stability to the system, mitigating risks associated with large-scale liquidations.

However, it is crucial to qualitatively assess each LST to determine its viability as collateral on the Liquity platform. Factors such as the stability of the LST, the reliability of the underlying staking protocol, and market acceptance must be considered. This assessment ensures that only robust and dependable LSTs are utilized, maintaining the integrity and resilience of the collateral system.

## 8 Appendix

Below, we present a stress scenario in which the price of ETH experiences a 70% decline over the course of half a year. The simulation is governed by the configuration of specific parameter values. We compare two distinct scenarios under the same seed: one with no minimum borrow rate and the other with a minimum borrow rate of 0.5%. The accompanying plots provide insight into how the system reacts in both scenarios.

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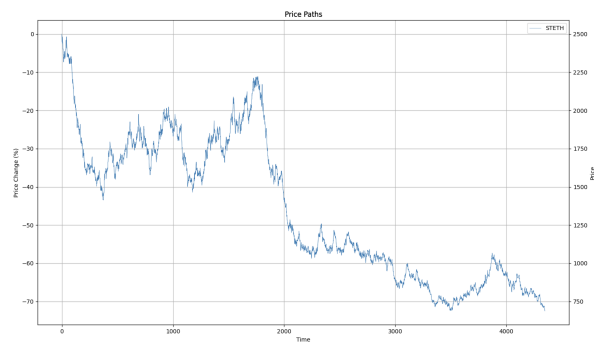


Figure 26: ETH Price Trajectory

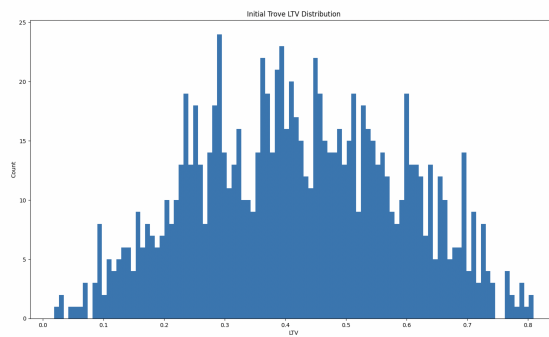


Figure 27: Initial Trove LTV Distribution

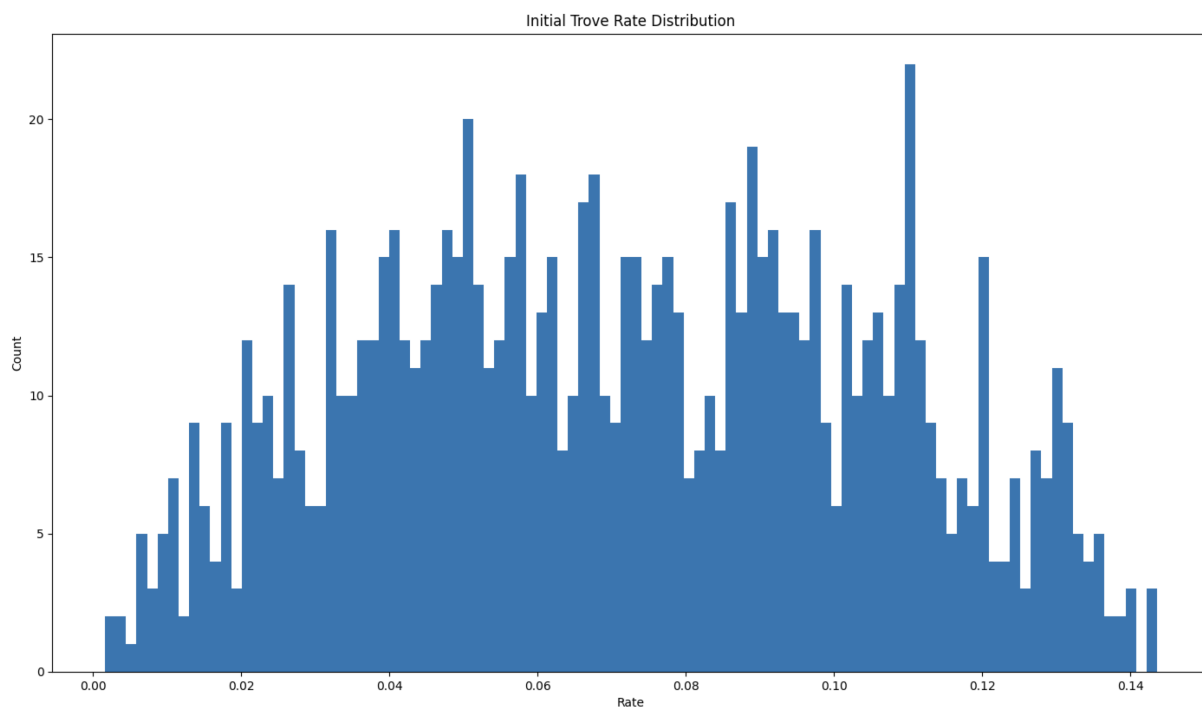


Figure 28: Initial Trove Rate Distribution

## No Minimum Borrow Rate

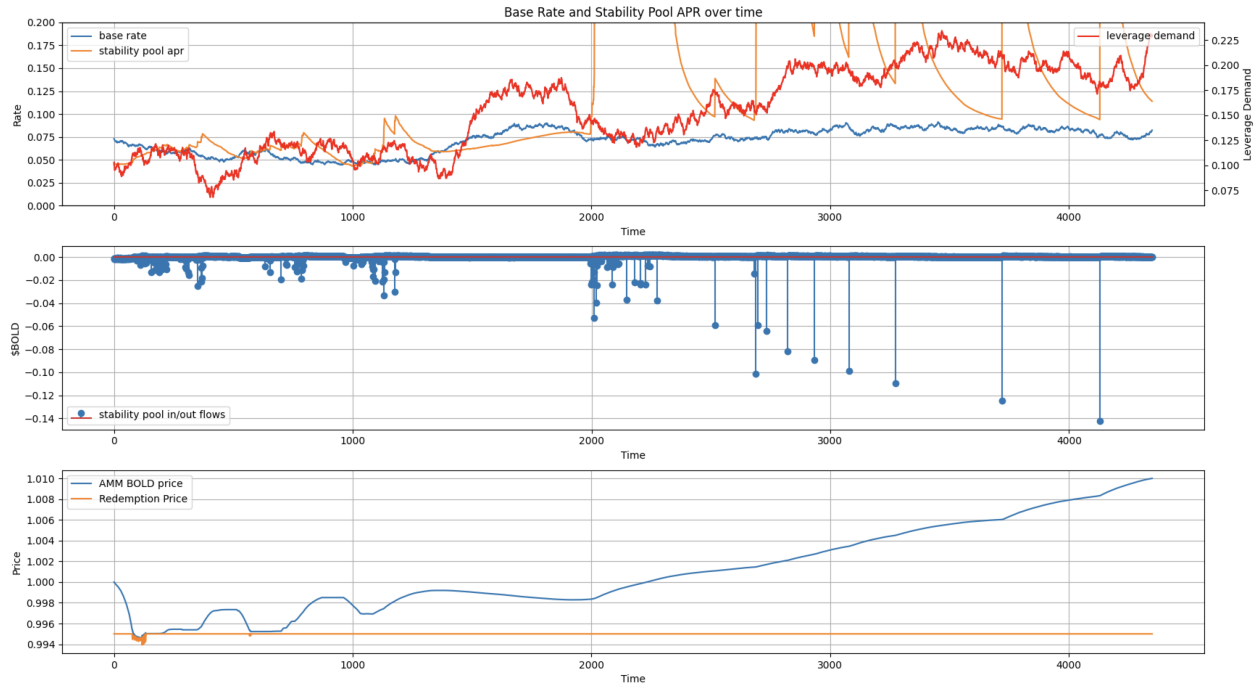


Figure 29: Base Rate, Stability Pool and Peg Dynamics Over Time

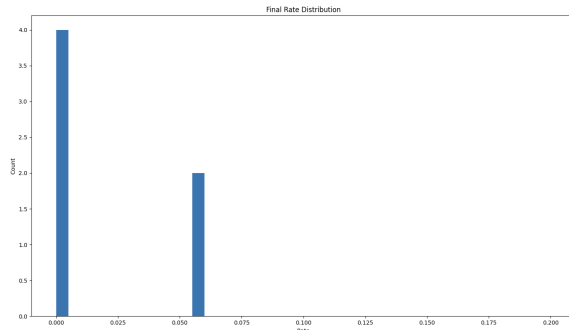


Figure 30: Final Rate Distribution

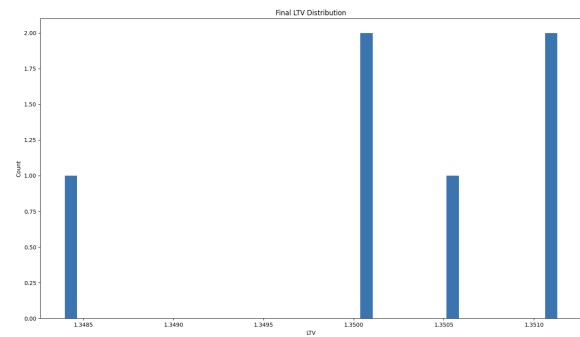


Figure 31: Final LTV Distribution

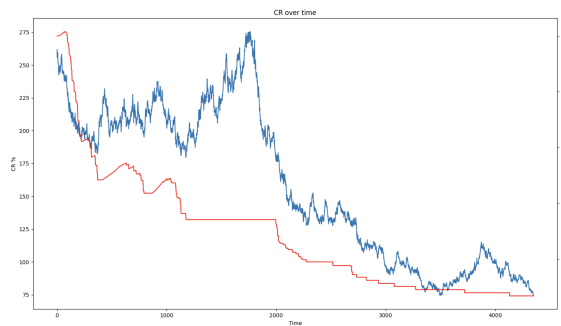


Figure 32: CR and Total Debt Over Time

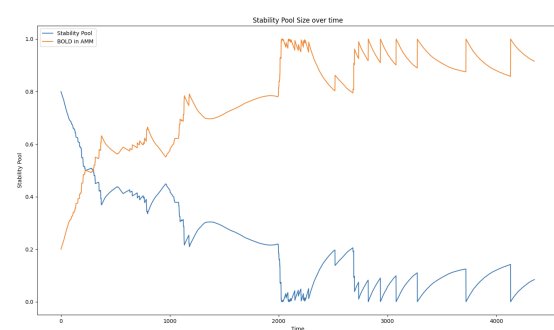


Figure 33: Stability Pool Size Over Time

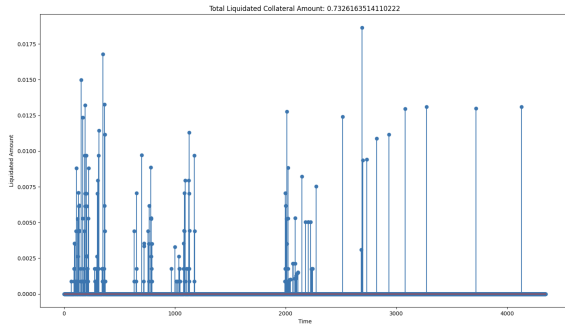


Figure 34: Total Liquidated Collateral Amount

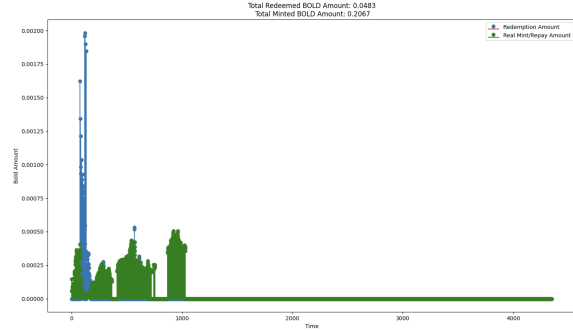


Figure 35: Total Minted/Redeemed BOLD Amount

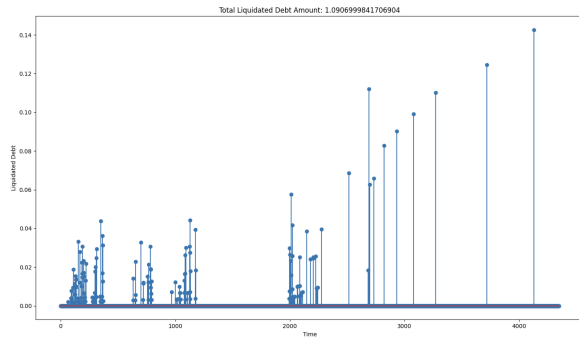


Figure 36: Total Burned Debt Amount

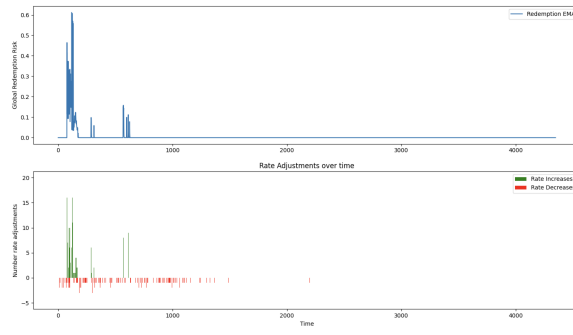


Figure 37: Rate Adjustments Over Time

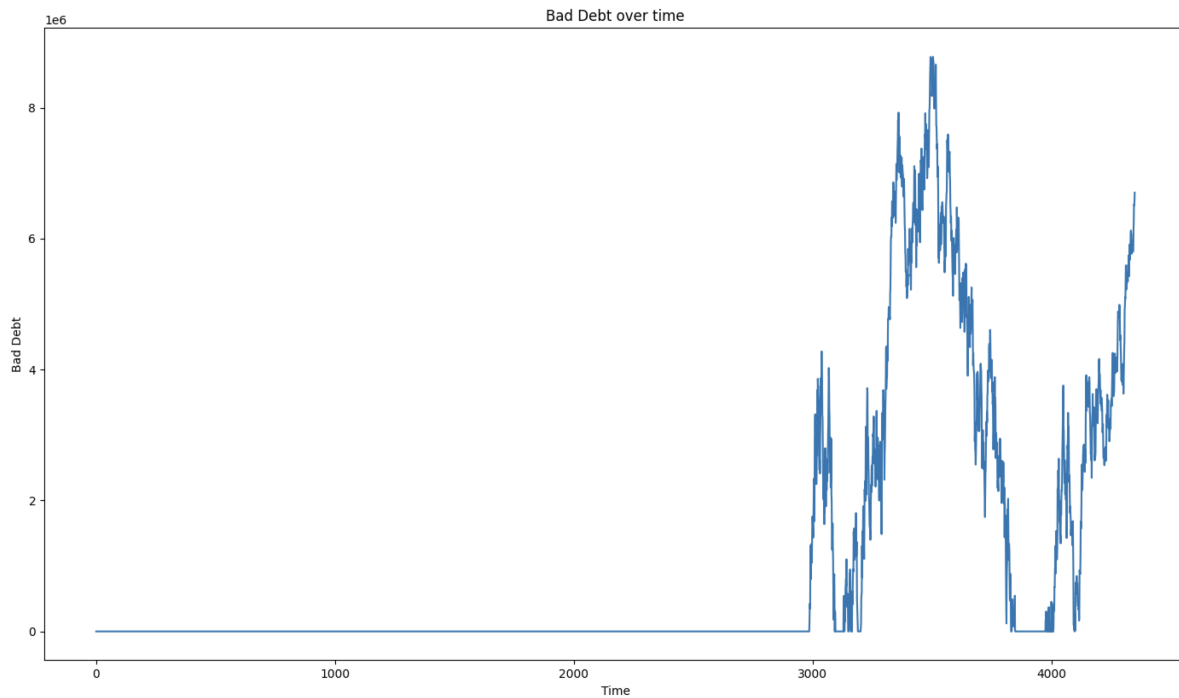


Figure 38: Bad Debt Over Time

## 0.5% Minimum Borrow Rate

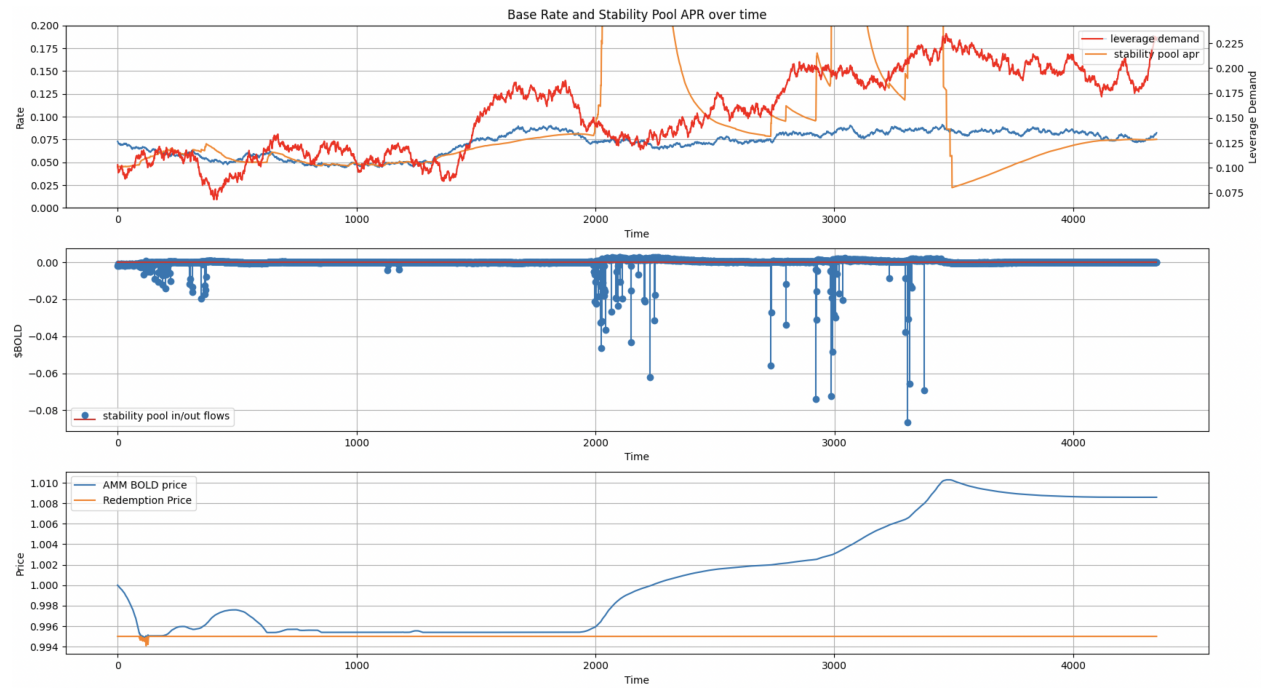


Figure 39: Base Rate, Stability Pool and Peg Dynamics Over Time

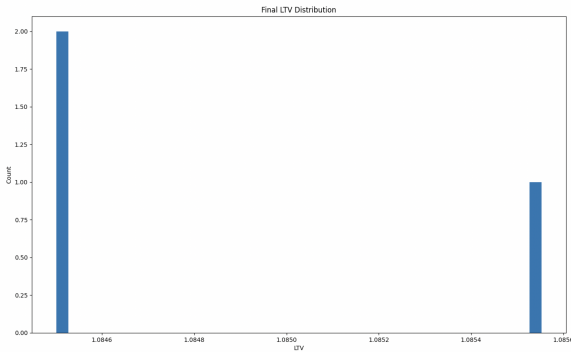


Figure 40: Final LTV Distribution

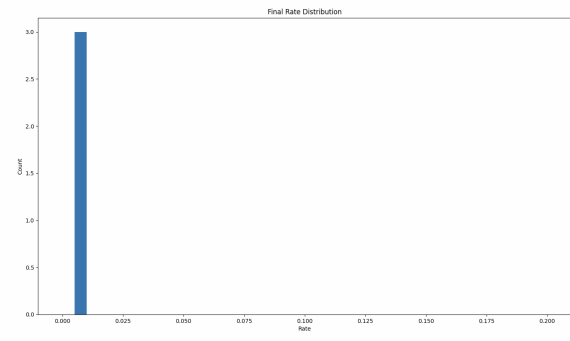


Figure 41: Final Rate Distribution

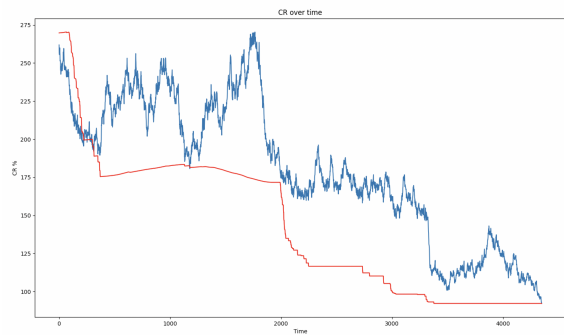


Figure 42: CR and Total Debt Over Time

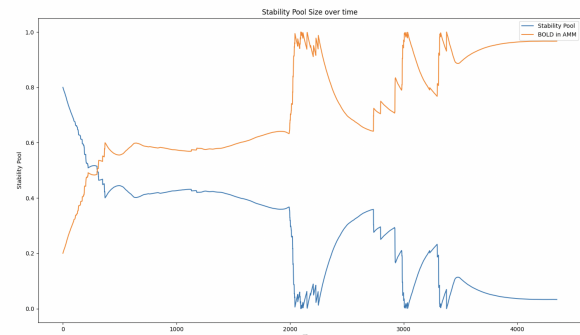


Figure 43: Stability Pool Size Over Time



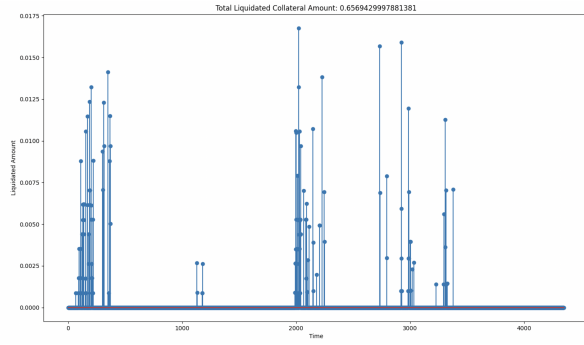


Figure 44: Total Liquidated Collateral Amount

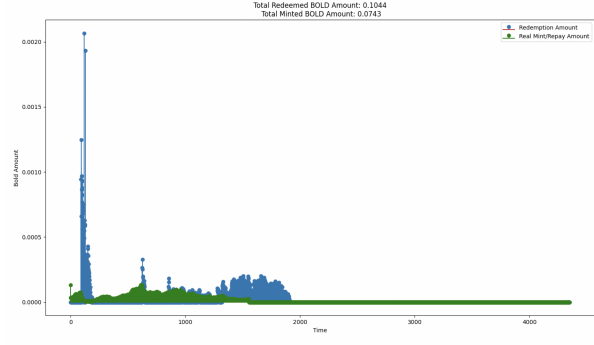


Figure 45: Total Minted/Redeemed BOLD Amount

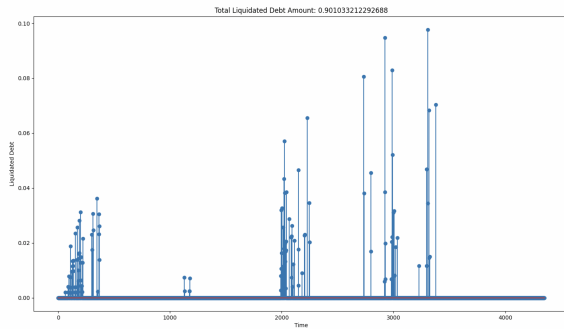


Figure 46: Total Burned Debt Amount

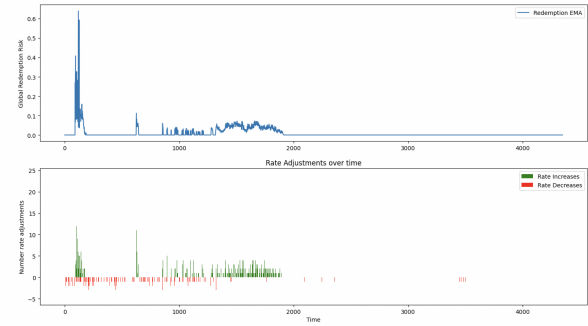


Figure 47: Rate Adjustments Over Time

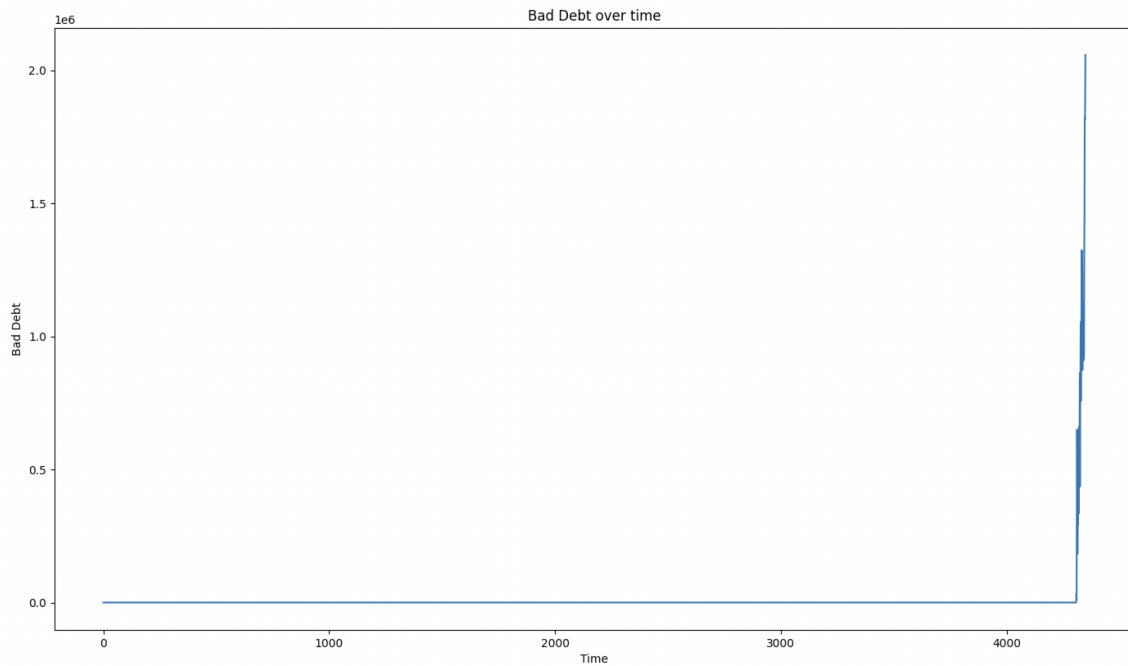


Figure 48: Bad Debt Over Time

## About Chaos Labs

Chaos Labs is a cloud-based platform that develops risk management and economic security tools for decentralized finance (DeFi) protocols. The platform leverages sophisticated and scalable simulations to stress test protocols in adverse and turbulent market conditions. By partnering with DeFi protocols, Chaos Labs aims to create innovative solutions that enhance the efficiency of DeFi marketplaces. The Chaos Labs team exhibits exceptional talent and represents diverse expertise, encompassing esteemed researchers, engineers, and security professionals. Chaos Labs has garnered its experience and skills from renowned organizations, including Google, Meta, Goldman Sachs, Instagram, Apple, Amazon, and Microsoft. Additionally, the team boasts members who have served in esteemed cyber-intelligence and security military units, further contributing to their unparalleled capabilities. You can explore our past and ongoing projects for customers like Aave, GMX, Benqi, dYdX, Uniswap, Maker, and more in the Research and Blog sections of our website.